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Design of a 400 Gb/s Silicon Photonics Receiver for IEEE 802.3 DR4 Datacenter Interconnects

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List of Abbreviations

Abbreviation	Meaning
APD	Avalanche photodiode
BOX	Buried oxide
CMOS	Complementary metal-oxide-semiconductor
CSPR	Carrier-to-signal power ratio
CW	Continuous wave
DR4	400GBASE-DR4 parallel single-mode fiber interface
DSP	Digital signal processing
EQE	External quantum efficiency
FCA	Free-carrier absorption
FDTD	Finite-difference time-domain
IQE	Internal quantum efficiency
NEP	Noise-equivalent power
OOK	On-off keying
PAM-4	Four-level pulse-amplitude modulation
PDL	Polarization-dependent loss
PIN	p-intrinsic-n photodiode
PINN	Physics-informed neural network
PML	Perfectly matched layer
RC	Resistance-capacitance
RF	Radio frequency
SOI	Silicon-on-insulator
SRH	Shockley-Read-Hall
SNR	Signal-to-noise ratio
SNI	Surface-normal illumination
SSBI	Signal-signal beating interference
TDD	Threading dislocation density
UTC	Uni-travelling-carrier

WG

Waveguide

List of Symbols

Symbol	Meaning	Unit
α	Absorption coefficient	cm^{-1}
$\tilde{n}$	Complex refractive index	—
n	Real part of refractive index	—
κ	Extinction coefficient	—
ω	Angular frequency	rad s^{-1}
μ_0	Permeability of free space	H m^{-1}
ε_0	Permittivity of free space	F m^{-1}
$\tilde{\varepsilon}$	Complex permittivity	F m^{-1}
$\tilde{k}$	Complex wave number	m^{-1}
c	Speed of light in vacuum	m s^{-1}
E	Electric field	V m^{-1}
H	Magnetic field	A m^{-1}
S	Poynting vector	W m^{-2}
G_{opt}	Optical generation rate	$\text{cm}^{-3} \text{s}^{-1}$
$\hbar$	Reduced Planck constant	J s
i_{ph}	Photocurrent	A
$\mathcal{R}$	Responsivity	A W^{-1}
P	Optical power	W
η_0	Impedance of free space	Ω
$\tilde{s}$	Complex envelope	—
ω_0	Carrier frequency	rad s^{-1}
c	DC carrier amplitude	—
$\tilde{u}$	Modulated signal amplitude	—
$CSPR$	Carrier-to-signal power ratio	dB
$E_{g,\text{ind}}$	Indirect bandgap energy	eV
$E_{g,\text{dir}}$	Direct bandgap energy	eV

Chapter 1

Design and Simulation of a Ge-on-Si Waveguide Photodetector for O-Band 400G Reception

1.1 Introduction and Chapter Scope

The relentless growth of data-centre interconnect capacity has driven the O-band silicon-photonics ecosystem toward 400G and beyond. IEEE 802.3 Clause 140 specifies 400GBASE-DR4 as a four-lane PAM-4 interface at 1310 nm, placing each lane at $106.25 \text{ Gbit s}^{-1}$ raw line rate and demanding a receiver-side photodetector that is simultaneously fast, efficient, and quiet. Meeting that specification on a CMOS-foundry platform is not merely a component selection problem: it requires a design methodology that connects photon absorption physics, material constraints, device topology, and electrode engineering into a single coherent argument.

This chapter develops that argument from first principles. The discussion is structured as a progressive narrowing: beginning with the electromagnetic foundations of light absorption in semiconductors, passing through the metrics that define a useful receiver device, surveying the landscape of competing architectures and materials, and arriving at a specific vertical n-i-p Ge-on-Si waveguide photodetector whose geometry, contact topology, and simulation strategy are each justified by the physics established along the way.

The chapter is organised as follows. Section 1.2 establishes the physics of photodetection, deriving the Beer-Lambert absorption law from Maxwell’s equations, developing the drift-diffusion carrier transport framework, and quantifying the recombination mechanisms that limit carrier collection. Section 1.3 defines the evaluation language, namely responsivity, bandwidth, dark current, noise, and the trade-offs among them, that governs every architectural and material choice that follows. Section 1.4 surveys photodetector device classes from the photoconductor through the avalanche photodiode and derives, rather than asserts,

the selection of the waveguide-integrated PIN junction for high-speed integrated links. Section 1.5 justifies germanium-on-silicon as the material platform, including its spectral suitability at 1310 nm, its CMOS integration path, and the threading-dislocation challenge that sets the dark-current floor. Section 1.6 reviews the experimental record of Ge-on-Si waveguide photodetectors critically and identifies the gap that motivates the U-shaped electrode approach adopted here. Section 1.7 presents the proposed device architecture in detail, including the vertical PIN layer stack, the adiabatic taper coupling strategy, the U-shaped electrode topology, and the geometry-to-performance design space mapping. Section 1.8 describes the multiphysics simulation strategy: FDTD optical modelling, CHARGE drift-diffusion solving, MATLAB post-processing, and the validation approach anchoring the simulation to the published benchmark. Section 1.9 closes with a synthesis of the key design insights and a transition to the results chapter.

1.2 Fundamentals of Photodetection

Photodetection begins with the transfer of optical energy to the semiconductor lattice. Understanding that process at the physics level is a prerequisite for rational device design, because every performance metric defined in section 1.3 traces back to the rate at which photons are absorbed, how the resulting carriers are transported, and what fraction is lost to recombination before reaching the device terminals.

1.2.1 Optical Absorption in Semiconductors

The starting point for all optical analysis in this chapter is the macroscopic form of Maxwell's equations in a non-magnetic, source-free medium. In SI units and in the frequency domain, obtained by assuming a time dependence $e^{-i\omega t}$, these read:

$$\nabla \times \tilde{\mathbf{E}}(\mathbf{r}, \omega) = i\omega\mu_0 \tilde{\mathbf{H}}(\mathbf{r}, \omega), \quad (1.1)$$

$$\nabla \times \tilde{\mathbf{H}}(\mathbf{r}, \omega) = -i\omega\tilde{\epsilon}(\omega) \tilde{\mathbf{E}}(\mathbf{r}, \omega), \quad (1.2)$$

where μ_0 is the permeability of free space and $\tilde{\epsilon}(\omega)$ is the complex permittivity. Taking the curl of eq. (1.1) and substituting eq. (1.2) yields the Helmholtz wave equation:

$$\nabla^2 \tilde{\mathbf{E}}(\mathbf{r}, \omega) + \tilde{k}^2(\omega) \tilde{\mathbf{E}}(\mathbf{r}, \omega) = \mathbf{0}, \quad (1.3)$$

where $\tilde{k}(\omega) = \omega\sqrt{\mu_0\tilde{\epsilon}(\omega)}$. Defining the *complex refractive index*:

$$\tilde{n}(\omega) \equiv \frac{c\tilde{k}(\omega)}{\omega} = n(\omega) + i\kappa(\omega), \quad (1.4)$$

the plane-wave solution propagating along $+\hat{z}$ is:

$$\tilde{\mathbf{E}}(z, \omega) = \mathbf{E}_0 \exp\left[i n(\omega) \frac{\omega z}{c}\right] \exp\left[-\kappa(\omega) \frac{\omega z}{c}\right]. \quad (1.5)$$

The time-averaged Poynting intensity $I(z) \propto |\tilde{\mathbf{E}}|^2$ decays as $\exp[-2\kappa\omega z/c]$, giving the Beer-Lambert absorption coefficient:

$$\alpha(\omega) = \frac{2\omega\kappa(\omega)}{c} = \frac{4\pi\kappa(\lambda)}{\lambda}, \quad (1.6)$$

so that the Beer-Lambert law:

$$I(z) = I_0 \exp[-\alpha(\lambda) z] \quad (1.7)$$

emerges as a corollary of eq. (1.3) rather than an independent empirical postulate. The absorbed power density follows directly:

$$\mathcal{P}_{\text{abs}}(\mathbf{r}) = \alpha(\lambda) I(\mathbf{r}). \quad (1.8)$$

Poynting-vector formulation of the generation rate.

The optical generation rate $G_{\text{opt}}(\mathbf{r})$ is more generally obtained from the time-averaged divergence of the Poynting vector. The divergence theorem applied to the time-averaged Poynting flux $\langle \mathbf{S} \rangle = \frac{1}{2} \text{Re}[\tilde{\mathbf{E}} \times \tilde{\mathbf{H}}^*]$ over a closed surface $\partial\Omega$ enclosing a volume Ω of absorbing material gives:

$$\int_{\Omega} G_{\text{opt}}(\mathbf{r}) \hbar\omega \, dV = - \oint_{\partial\Omega} \langle \mathbf{S} \rangle \cdot \hat{n} \, dA, \quad (1.9)$$

so that the local volumetric generation rate is the Poynting-flux sink:

$$G_{\text{opt}}(\mathbf{r}) = -\frac{\nabla \cdot \langle \mathbf{S}(\mathbf{r}) \rangle}{\hbar\omega} = \frac{\omega\varepsilon_0 \text{Im}[\tilde{\varepsilon}(\mathbf{r})] |\tilde{\mathbf{E}}(\mathbf{r})|^2}{2\hbar\omega}. \quad (1.10)$$

This reduces identically to eq. (1.15) under the plane-wave approximation, but is the operationally correct definition when the three-dimensional field contains interference terms arising from modal coupling, metal reflections, or end-facet diffraction, all of which are present in the adiabatic taper region of the device studied here. The FDTD solver employed in section 1.8.1 evaluates G_{opt} from eq. (1.10); the scalar Beer-Lambert law of eq. (1.7) serves only as the analytical validation baseline.

Square-law photodetection and the domain of linear operation.

For a quasi-monochromatic field at carrier frequency ω_0 with slowly varying complex envelope $\tilde{s}(t)$:

$$\tilde{\mathbf{E}}(t) = \frac{1}{2} \tilde{s}(t) e^{-i\omega_0 t} \hat{\mathbf{e}} + \text{c.c.}, \quad (1.11)$$

the time-averaged Poynting intensity at the detector facet is $I(t) = |\tilde{s}(t)|^2/(2\eta_0)$, where $\eta_0 = \sqrt{\mu_0/\varepsilon_0}$. The photocurrent produced by the device is therefore:

$$i_{\text{ph}}(t) = \mathcal{R} \cdot P(t) = \mathcal{R} \cdot \frac{|\tilde{s}(t)|^2}{2\eta_0}, \quad (1.12)$$

which is *linear in optical power* $P(t)$ but *quadratic in optical field amplitude* $|\tilde{s}(t)|$. When the field contains both a data-bearing modulated component $\tilde{u}(t)$ and a residual DC carrier c , $\tilde{s}(t) = \tilde{u}(t) + c$, the photocurrent expands as:

$$i(t) = \mathcal{R} \left[\underbrace{c^2}_{\text{DC bias}} + \underbrace{2c \text{Re}\{\tilde{u}(t)\}}_{\text{linear signal term}} + \underbrace{|\tilde{u}(t)|^2}_{\text{SSBI}} \right], \quad (1.13)$$

where SSBI denotes signal-signal beating interference. The linear approximation $i_{\text{ph}} \approx \mathcal{R} \cdot P_{\text{opt}}$ is valid when the carrier-to-signal power ratio $\text{CSPR} \equiv c^2 / \langle |\tilde{u}|^2 \rangle \gg 1$, so that SSBI is negligible relative to the cross-beat term.

Integrating eq. (1.8) through an absorber of length L gives the absorbed fraction:

$$\mathcal{A}(L) = 1 - \exp[-\alpha(\lambda) L]. \quad (1.14)$$

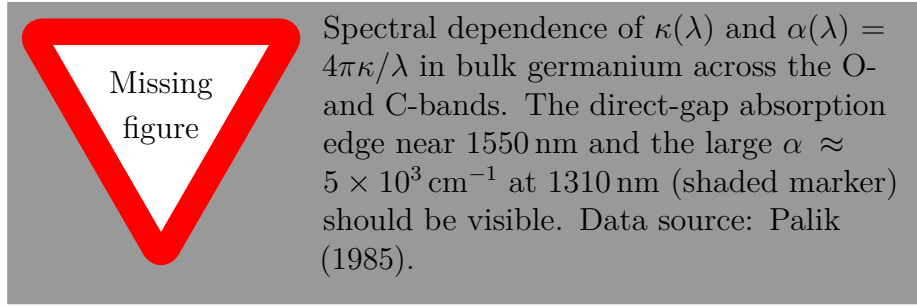


Figure 1.1: Spectral absorption coefficient $\alpha(\lambda)$ and extinction coefficient $\kappa(\lambda)$ of bulk germanium, derived via eq. (1.6) from tabulated optical constants. The large α at 1310 nm reflects direct-gap absorption at the Γ -point discussed in section 1.5.

1.2.2 Carrier Photogeneration and Transport

The absorbed photon creates an electron-hole pair. Whether that pair is swept to the contacts and contributes to the photocurrent, or instead recombines within the absorber and is lost, depends on the interplay of the electric field, the carrier mobilities, and the defect density of the semiconductor.

The volumetric optical generation rate $G_{\text{opt}}(\mathbf{r})$ quantifies the number of electron-hole pairs created per unit volume per unit time:

$$G_{\text{opt}}(\mathbf{r}) = \frac{\mathcal{P}_{\text{abs}}(\mathbf{r})}{\hbar\omega} = \frac{\alpha(\lambda) I(\mathbf{r})}{\hbar\omega} \quad [\text{cm}^{-3} \text{s}^{-1}]. \quad (1.15)$$

Because $I(\mathbf{r})$ decays exponentially along the propagation direction via eq. (1.7), G_{opt} is spatially non-uniform: it is concentrated near the input facet of the absorber and falls off with a characteristic length $\alpha^{-1} \approx 4.8 \mu\text{m}$ at 1310 nm.

Drift–diffusion transport.

The self-consistent carrier distribution and electrostatic field are governed by the coupled Poisson and drift-diffusion equations [1], [2]. Poisson’s equation is:

$$\nabla \cdot [\varepsilon(\mathbf{r}) \nabla \phi(\mathbf{r}, t)] = -q[p - n + N_D^+ - N_A^-], \quad (1.16)$$

and the electron and hole current densities in the drift-diffusion approximation are:

$$\mathbf{J}_n = q\mu_n n \mathbf{E} + qD_n \nabla n, \quad (1.17)$$

$$\mathbf{J}_p = q\mu_p p \mathbf{E} - qD_p \nabla p, \quad (1.18)$$

where the Einstein relations $D_{n,p} = \mu_{n,p}kT/q$ hold for non-degenerate statistics. The general, time-dependent carrier continuity equations are:

$$\frac{\partial n}{\partial t} = \frac{1}{q} \nabla \cdot \mathbf{J}_n - U_{\text{net}} + G_{\text{opt}}, \quad (1.19)$$

$$\frac{\partial p}{\partial t} = -\frac{1}{q} \nabla \cdot \mathbf{J}_p - U_{\text{net}} + G_{\text{opt}}, \quad (1.20)$$

where $U_{\text{net}} = U_{\text{SRH}} + U_{\text{Auger}} + U_{\text{rad}}$. Under *steady-state* CW illumination, $\partial/\partial t \equiv 0$:

$$\frac{1}{q} \nabla \cdot \mathbf{J}_n = U_{\text{net}} - G_{\text{opt}}, \quad (1.21)$$

$$\frac{1}{q} \nabla \cdot \mathbf{J}_p = -(U_{\text{net}} - G_{\text{opt}}). \quad (1.22)$$

The complete self-consistent system, from eq. (1.16) through eq. (1.22), is the foundational PDE set solved numerically in both the commercial finite-element solver and the PINN framework discussed in section 1.8.4.

The central design consequence of the drift-diffusion picture is the distinction between drift-dominated and diffusion-dominated carrier motion. Drift transport, which is field-driven and fast, requires a sufficiently large reverse-bias field to sweep carriers across the intrinsic region before they can recombine. Diffusion transport, which is concentration-gradient-driven and inherently slower, dominates in the quasi-neutral regions outside the depletion zone and is the primary cause of slow response tails in devices with incomplete or absent depletion. A reverse-biased PIN junction suppresses diffusion by confining photogeneration to the depleted region, making drift the dominant transport mechanism and the carrier transit time the key speed parameter.

Analytical photocurrent transfer function.

In the frequency domain, neglecting diffusion and assuming a uniform electric field at saturation value throughout $z \in [0, W_i]$, the Fourier amplitude of the terminal current

density admits a closed-form expression:

$$J_t(\omega) = \frac{qG_0}{\alpha W_i} \frac{hc}{q\lambda} \eta(1 - R_{\text{refl}}) P_{\text{opt}}(\omega) \times \left\{ \frac{e^{-\alpha W_i} - 1}{\alpha W_i(\alpha W_i - j\omega\tau_{\text{tr},h})} + e^{-\alpha W_i} \frac{e^{j\omega\tau_{\text{tr},h}} - 1}{j\omega\tau_{\text{tr},h}(\alpha W_i - j\omega\tau_{\text{tr},h})} + \frac{1 - e^{-\alpha W_i}}{\alpha W_i(j\omega\tau_{\text{tr},e} + \alpha W_i)} + \frac{1 - e^{j\omega\tau_{\text{tr},e}}}{j\omega\tau_{\text{tr},e}(j\omega\tau_{\text{tr},e} + \alpha W_i)} \right\}, \quad (1.23)$$

where $\tau_{\text{tr},e} = W_i/v_{\text{d,sat},e}$ and $\tau_{\text{tr},h} = W_i/v_{\text{d,sat},h}$ are the electron and hole transit times. In the limits $\omega\tau_{\text{tr},h} \ll 1$ and $\alpha W_i \gg 1$, eq. (1.23) reduces to the rectangular-pulse sinc-function roll-off yielding the bandwidth formula of eq. (1.40).

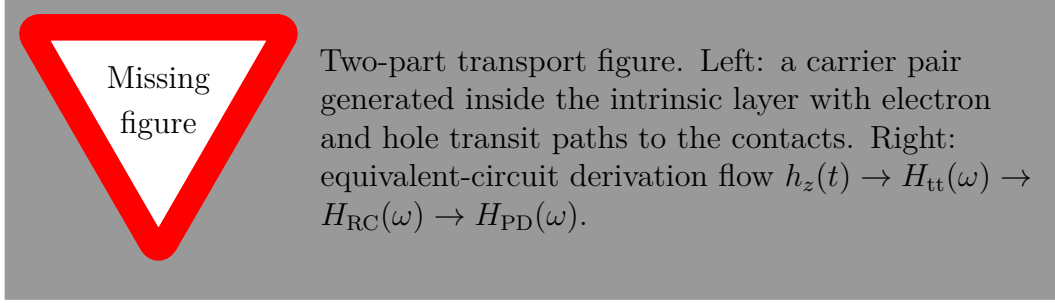


Figure 1.2: Microscopic-to-circuit derivation path for the small-signal transfer function. The transit-time response originates from the finite carrier flight time inside the intrinsic Ge region, while the RC response originates from the external electrical loading seen by the junction.

Derivation of the factorised small-signal transfer function.

Consider an electron-hole pair generated at depth $z \in [0, W_i]$ inside the intrinsic layer. Under the uniform-field, velocity-saturated approximation, the electron and hole transit times are:

$$\tau_e(z) = \frac{z}{v_{\text{d,sat},e}}, \quad \tau_h(z) = \frac{W_i - z}{v_{\text{d,sat},h}}. \quad (1.24)$$

By the Ramo-Shockley theorem, the terminal current induced by each carrier is non-zero only while the carrier is moving, giving the local impulse response:

$$h_z(t) = \frac{qv_{\text{d,sat},e}}{W_i} u(t) u[\tau_e(z) - t] + \frac{qv_{\text{d,sat},h}}{W_i} u(t) u[\tau_h(z) - t], \quad (1.25)$$

where $u(t)$ is the Heaviside function. Fourier transformation yields:

$$H_z(\omega) = \frac{q}{W_i} \left[\frac{1 - e^{-j\omega\tau_e(z)}}{j\omega} + \frac{1 - e^{-j\omega\tau_h(z)}}{j\omega} \right]. \quad (1.26)$$

If the optical generation is approximately uniform across the depletion depth, the transit response follows from averaging eq. (1.26) over z :

$$H_{\text{tt}}(\omega) \propto \int_0^{W_i} H_z(\omega) \frac{dz}{W_i}. \quad (1.27)$$

The integral evaluates to a weighted sum of electron and hole sinc terms. In the practically relevant limit where the slower hole branch dominates the 3 dB roll-off, the envelope reduces to the single-sinc form used in eq. (1.44). The electrical loading is then appended through the junction capacitance and total series resistance. For a current source driving C_j into $R_{\text{tot}} = R_s + R_L$:

$$H_{\text{RC}}(\omega) = \frac{1}{1 + j\omega R_{\text{tot}} C_j} = \frac{1}{1 + j\omega/\omega_{\text{RC}}}, \quad (1.28)$$

with $\omega_{\text{RC}} = (R_{\text{tot}} C_j)^{-1}$. The total response is the product:

$$H_{\text{PD}}(\omega) = H_{\text{tt}}(\omega) H_{\text{RC}}(\omega), \quad (1.29)$$

which is the physical content hidden inside the compact form of eq. (1.44): transport determines the sinc-like envelope, while the external electrical boundary condition determines the low-pass pole.

Scharfetter-Gummel discretisation and numerical stability.

The finite-element discretisation of eqs. (1.17) and (1.18) on a non-uniform mesh faces a well-known numerical instability when the local Péclet number $\text{Pe} = q|\mathbf{E}|\Delta x/(2kT)$ exceeds unity. The Scharfetter-Gummel scheme approximates the current between adjacent mesh nodes i and $i + 1$ as:

$$J_{n,i+1/2} = \frac{qD_n}{\Delta x} \left[B\left(\frac{q\Delta\phi}{kT}\right) n_{i+1} - B\left(-\frac{q\Delta\phi}{kT}\right) n_i \right], \quad (1.30)$$

where $B(u) = u/(e^u - 1)$ is the Bernoulli function and $\Delta\phi = \phi_{i+1} - \phi_i$. This scheme recovers centred differences for $|u| \ll 1$ (low-field) and upstream weighting for $|u| \gg 1$ (high-field), providing second-order accuracy and unconditional positivity across the 30 kV cm⁻¹ field regime of the present device.

1.2.3 Recombination and Loss Mechanisms

Three recombination mechanisms are quantitatively relevant in the germanium absorber [1], [3].

Shockley-Read-Hall (SRH) recombination, mediated by deep-level trap states, is:

$$U_{\text{SRH}} = \frac{np - n_i^2}{\tau_p(n + n_1) + \tau_n(p + p_1)}, \quad (1.31)$$

where $n_1 = n_i \exp[(E_t - E_i)/kT]$ and $p_1 = n_i \exp[(E_i - E_t)/kT]$. Within the fully depleted intrinsic region at the midgap trap condition $E_t = E_i$, eq. (1.31) reduces to:

$$U_{\text{SRH,dep}} \approx \frac{n_i}{\tau_n + \tau_p} \equiv \frac{n_i}{2\tau_0}, \quad (1.32)$$

where $\tau_0 = (\tau_n + \tau_p)/2$. SRH is the dominant mechanism in Ge-on-Si because the threading dislocations formed during heteroepitaxial growth act as efficient midgap trap centres.

Auger recombination follows $U_{\text{Auger}} = C_n n(np - n_i^2) + C_p p(np - n_i^2)$.

Radiative recombination is modelled as:

$$U_{\text{rad}} = r_{\text{opt}}(np - n_i^2), \quad (1.33)$$

where $r_{\text{opt}}^{\text{Ge}} \approx 6.4 \times 10^{-14} \text{ cm}^3 \text{ s}^{-1}$ at 300 K, approximately four orders of magnitude smaller than for a direct-gap III-V of comparable bandgap. In the fully depleted intrinsic region, where $np \approx n_i^2$, the radiative rate vanishes identically and eq. (1.32) constitutes the sole generation mechanism. Simulated generation and recombination rates confirm that SRH exceeds Auger by more than two orders of magnitude and radiative recombination by more than five orders of magnitude across $0 \leq |V_{\text{bias}}| \leq 3 \text{ V}$. This justifies the single-parameter calibration strategy of section 1.8.2, in which only τ_0 is adjusted to reproduce I_d .

Surface recombination at the Ge/SiO₂ and Ge/Si interfaces constitutes a separate loss channel characterised by the surface recombination velocity v_s . Its contribution to the dark current is proportional to the mesa perimeter P_j rather than the junction area A_j , as formalised in eqs. (1.51) and (1.52). As device dimensions shrink to improve bandwidth, the area-to-perimeter ratio A_j/P_j decreases and surface recombination becomes increasingly significant, setting a practical lower bound on the achievable dark current density that surface passivation, for example through Ge_xO_y or Al₂O₃, must address.

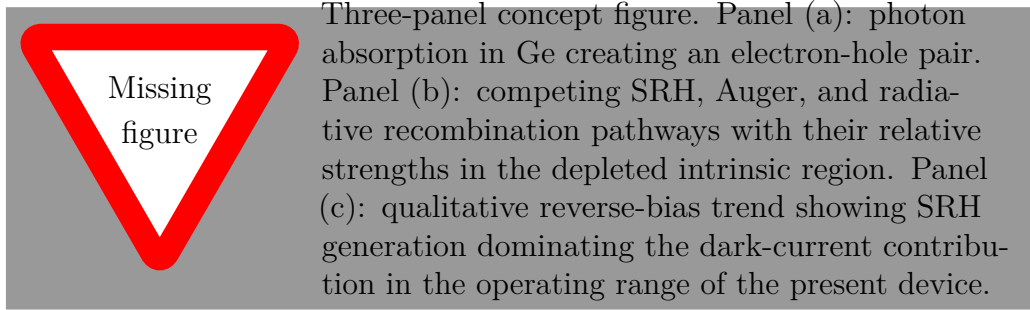


Figure 1.3: Carrier generation and recombination picture used throughout the electrical model. In the fully depleted intrinsic Ge layer, SRH generation remains the dominant dark-current mechanism, which justifies calibrating the device primarily through the lifetime parameter τ_0 .

1.3 Performance Metrics and Design Trade-offs

The transport and absorption physics of section 1.2 support a set of engineering metrics that define what a useful photodetector must accomplish and at what cost. These metrics are not independent: they form a multi-dimensional trade-off space in which improving one quantity typically degrades another. Every architectural and material choice in the sections that follow is justifiable using the definitions established here.

1.3.1 Responsivity and Quantum Efficiency

The *external responsivity* $\mathcal{R}$ is:

$$\mathcal{R} \equiv \frac{I_{\text{ph}}}{P_{\text{opt}}} \quad [\text{A W}^{-1}], \quad (1.34)$$

with theoretical upper bound:

$$\mathcal{R}_{\text{max}} = \frac{q\lambda}{hc} \approx 1.057 \text{ A W}^{-1} \quad \text{at } 1310 \text{ nm}. \quad (1.35)$$

The external quantum efficiency (EQE) η_{ext} and internal quantum efficiency (IQE) η_{int} are:

$$\eta_{\text{ext}} = \mathcal{R} \cdot \frac{hc}{q\lambda}, \quad (1.36)$$

$$\eta_{\text{int}} = \frac{I_{\text{ph}}/q}{P_{\text{abs}}/\hbar\omega}. \quad (1.37)$$

In the fully depleted intrinsic region at saturation field, the drift transit time $\tau_{\text{tr}} = W_i/v_{\text{d,sat},h}$ is much shorter than the SRH lifetime τ_0 , giving a carrier collection efficiency $\xi \approx 1 - \tau_{\text{tr}}/\tau_0 \approx 1$. The more general form of the EQE chain is:

$$\eta_{\text{ext}} = (1 - R_{\text{refl}}) \xi \eta_c (1 - e^{-\alpha(\lambda)L}), \quad (1.38)$$

where R_{refl} is the Si/Ge entry-facet reflectance and η_c is the coupling efficiency from the silicon waveguide into the Ge absorber mode. Real devices fall below $\mathcal{R}_{\text{max}}$ because of coupling loss ($\eta_c < 1$), incomplete absorption ($e^{-\alpha L} > 0$), and residual carrier recombination before collection ($\xi < 1$).

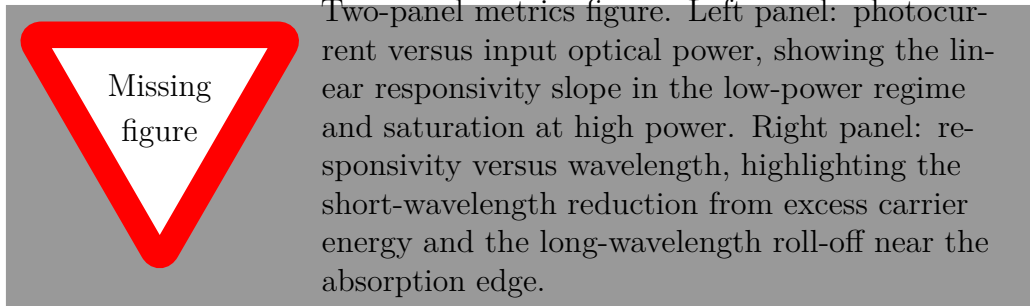


Figure 1.4: Responsivity viewed in both its power-domain and spectral forms. The low-power slope of the I - P curve defines $\mathcal{R}$, whereas the wavelength dependence determines the useful operating window around the O-band.

1.3.2 Bandwidth: Transit-Time and RC Limits

The frequency response is limited by two physically distinct mechanisms. In the absence of deliberate inductive peaking, the combined baseline bandwidth is [4]:

$$f_{3\text{dB}} = \left(\frac{1}{f_{\text{tt}}^2} + \frac{1}{f_{\text{RC}}^2} \right)^{-1/2}. \quad (1.39)$$

Transit-time limit.

With hole saturation velocity $v_{\text{d,sat},h} \approx 4 \times 10^6 \text{ cm s}^{-1}$ and electron saturation velocity $v_{\text{d,sat},e} \approx 6 \times 10^6 \text{ cm s}^{-1}$, the transit-time-limited 3 dB bandwidth from Fourier analysis of a rectangular current impulse response is:

$$f_{\text{tt}} = \frac{0.45 v_{\text{d,sat},h}}{W_i}, \quad (1.40)$$

where the coefficient 0.45 arises from the sinc-like spectral roll-off and reflects the asymmetric electron-hole velocity ratio $v_{\text{d,sat},e}/v_{\text{d,sat},h} \approx 1.5$. For $W_i = 350 \text{ nm}$: $f_{\text{tt}} \approx 51 \text{ GHz}$. This value is a geometry-screening lower bound rather than the absolute transport ceiling. The full two-carrier response of eq. (1.23) is broader because electrons and holes induce terminal current simultaneously, and the experimentally relevant RF response further benefits from on-electrode inductance L_p . For the 350 nm intrinsic region, the benchmark-calibrated transport ceiling is in the 110 GHz to 120 GHz class.

RC limit.

The junction capacitance of the fully depleted intrinsic region is:

$$C_j = \frac{\varepsilon_0 \varepsilon_{\text{Ge}} A_j}{W_i}, \quad (1.41)$$

where $\varepsilon_{\text{Ge}} \approx 16.2$ and $A_j = W_m \times L$. The RC-limited bandwidth is:

$$f_{\text{RC}} = \frac{1}{2\pi R_{\text{tot}} C_j}, \quad (1.42)$$

where $R_{\text{tot}} = R_s + R_L$ and $R_L = 50 \Omega$.

Optimal intrinsic-layer thickness.

Setting $f_{\text{tt}}(W_i^*) = f_{\text{RC}}(W_i^*)$ and substituting eqs. (1.40) to (1.42) gives:

$$W_i^* = \sqrt{0.9\pi R_{\text{tot}} \varepsilon_0 \varepsilon_{\text{Ge}} v_{\text{d,sat},h} A_j}. \quad (1.43)$$

Substituting $R_{\text{tot}} = 60 \Omega$, $\varepsilon_{\text{Ge}} = 16.2$, $v_{\text{d,sat},h} = 4 \times 10^6 \text{ cm s}^{-1}$, and $A_j = 6 \mu\text{m}^2$ gives $W_i^* \approx 310 \text{ nm}$, within 11 % of the chosen $W_i = 350 \text{ nm}$, confirming that the device is positioned near the bandwidth optimum. The expression eq. (1.43) reveals a fundamental scaling law: since $W_i^* \propto \sqrt{A_j}$, shrinking the junction area by a factor of four allows W_i^* to be halved, raising the balanced bandwidth at the new optimum by approximately the same factor.

The small-signal photocurrent transfer function is well-approximated by the factorised form:

$$H_{\text{PD}}(f) = \text{sinc}\left(\frac{f}{f_{\text{tt}}}\right) \cdot \frac{1}{1 + i f/f_{\text{RC}}}, \quad (1.44)$$

where $\text{sinc}(x) = \sin(\pi x)/(\pi x)$. For the benchmark-calibrated U-shaped device, the electrical sub-network is more accurately represented by a series branch $Z_s(\omega) = R_s + j\omega L_p$ driving a terminated capacitive load $Z_L(\omega) = (R_L^{-1} + j\omega C_p)^{-1}$ while the reverse-biased junction impedance is $Z_j(\omega) \approx (j\omega C_j)^{-1}$. Current division then gives the peaked electrical transimpedance:

$$H_{\text{elec,pk}}(\omega) = \frac{Z_j(\omega) Z_L(\omega)}{Z_j(\omega) + Z_s(\omega) + Z_L(\omega)}, \quad (1.45)$$

and the full benchmark-calibrated detector response becomes:

$$H_{\text{PD,pk}}(\omega) = H_{\text{tt}}(\omega) H_{\text{elec,pk}}(\omega). \quad (1.46)$$

Equation (1.44) is used for fast trade-off screening; eq. (1.46) is used when projecting the validated single-U device toward the 100 GHz-class operating regime.

Series resistance and RC time constant.

The series resistance R_s in the vertical n-i-p architecture is dominated by the lateral conduction path through the p^{++} -Si platform:

$$R_{s,\text{anode}} \approx \frac{\rho_{\text{Si}}}{t_{\text{Si}}} \cdot \frac{\ell_a}{w_a}, \quad (1.47)$$

where $t_{\text{Si}} = 220 \text{ nm}$ is the platform thickness. The RC time constant is:

$$\tau_{\text{RC}} = R_s C_j, \quad (1.48)$$

from which $f_{\text{RC}} = (2\pi\tau_{\text{RC}})^{-1}$.

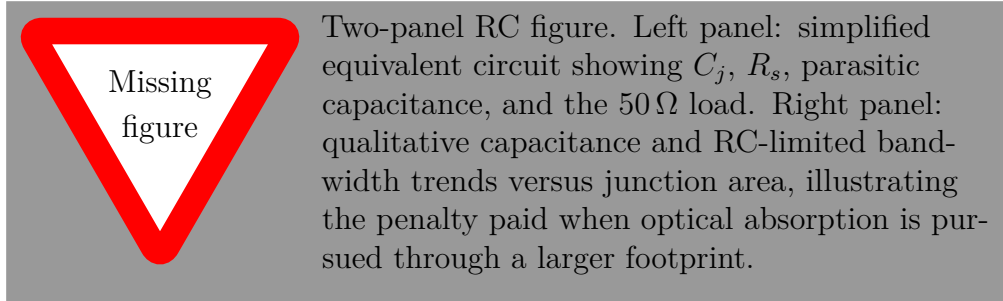


Figure 1.5: Capacitance-resistance interpretation of the vertical PIN photodetector. The key RC penalty comes from simultaneously increasing the depleted junction area and the lateral access resistance, which is why layout and footprint must be optimised together.

1.3.3 Noise, Sensitivity, and Figures of Merit

Dark Current

Dark current I_d flows through the reverse-biased junction in the absence of optical excitation and sets the noise floor through eq. (1.55). Three generation-recombination mechanisms

contribute [1]. The *SRH generation* current density within the depleted intrinsic layer of thickness W_i follows from eq. (1.32):

$$J_{\text{SRH}} = q U_{\text{SRH,dep}} W_i = \frac{q n_i W_i}{2\tau_0}. \quad (1.49)$$

Minority-carrier diffusion from the quasi-neutral contact regions contributes:

$$J_{\text{diff}} = \frac{q D_p n_i^2}{L_p N_D} + \frac{q D_n n_i^2}{L_n N_A}, \quad (1.50)$$

where $L_p = \sqrt{D_p \tau_p}$ and $L_n = \sqrt{D_n \tau_n}$. *Interface recombination* at the Ge/Si heterojunction contributes:

$$J_{\text{surf}} \approx \frac{q n_i v_s}{2}. \quad (1.51)$$

The total dark current, including both bulk and surface (perimeter) contributions, is:

$$I_d = A_j (J_{\text{SRH}} + J_{\text{diff}}) + P_j q n_i v_s / 2, \quad (1.52)$$

where $P_j = 2(W_m + L)$ is the junction mesa perimeter. This perimeter scaling has a direct consequence for miniaturisation: as A_j is reduced to decrease C_j and improve bandwidth, the area-to-perimeter ratio A_j/P_j decreases, increasing the fractional contribution of the surface term.

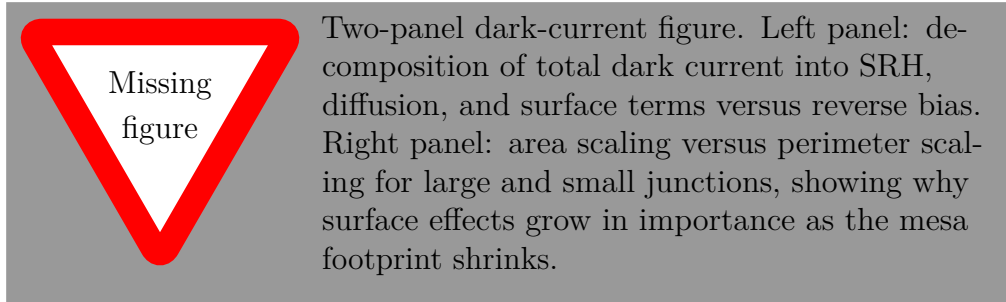


Figure 1.6: Dark-current mechanisms of the reverse-biased Ge-on-Si PIN. The present device remains bulk-SRH dominated at the chosen footprint, but perimeter-driven surface generation becomes progressively more important as the junction is aggressively miniaturised.

Noise Spectral Densities and Noise-Equivalent Power

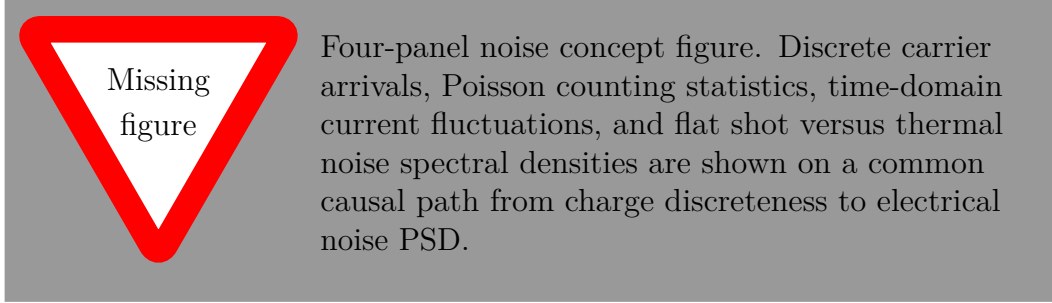


Figure 1.7: Figure-first explanation of noise origin. Shot noise is not an abstract fitting parameter but the continuum limit of the counting fluctuations of discrete carriers arriving at the terminal.

Origin of shot noise from carrier counting statistics.

The physical origin of shot noise is the randomness of discrete carrier arrivals. Let N_T denote the number of carriers collected during an observation interval T . If the arrivals are statistically independent, then N_T is Poisson distributed with mean $\mathbb{E}[N_T] = IT/q$ and variance $\text{Var}[N_T] = IT/q$. The collected charge is $Q_T = qN_T$, so:

$$\text{Var}[Q_T] = qIT. \quad (1.53)$$

Dividing by T^2 converts charge variance to variance of the time-averaged current:

$$\text{Var}[\bar{I}_T] = \frac{qI}{T}. \quad (1.54)$$

Passing from the finite-time variance to the one-sided continuous spectral density yields the standard result $S_{\text{shot}} = 2qI$.

Shot noise arises from the quantum-mechanical discreteness of charge carriers:

$$S_{\text{shot}}(f) = 2q(I_d + I_{\text{ph}}) \quad [\text{A}^2 \text{Hz}^{-1}]. \quad (1.55)$$

Johnson (thermal) noise from the load resistance contributes:

$$S_{\text{thermal}}(f) = \frac{4kT}{R_L} \quad [\text{A}^2 \text{Hz}^{-1}]. \quad (1.56)$$

Integrating over the detection bandwidth gives the total photodetector noise variance:

$$\sigma_n^2 = i_n^2 B_e \quad [\text{A}^2], \quad (1.57)$$

where $i_n^2 = S_{\text{shot}}$. The *noise-equivalent power* (NEP) and *specific detectivity* D^* are:

$$\text{NEP} = \frac{\sqrt{S_{\text{shot}} + S_{\text{thermal}}}}{\mathcal{R}} \quad [\text{W}/\sqrt{\text{Hz}}], \quad (1.58)$$

$$D_{\text{shot}}^* = \frac{\mathcal{R}\sqrt{A_j}}{\sqrt{2qI_d}}. \quad (1.59)$$

Generation-recombination noise from a trap ensemble with characteristic lifetime τ_{gr} has a Lorentzian form that falls below relevance in the tens-of-GHz band of interest. Optical-source noise enters as:

$$S_{\text{RIN}}(f) = \mathcal{R}^2 P_{\text{opt}}^2 \text{RIN}, \quad (1.60)$$

and can become important at the link level even when the detector itself is quiet. By contrast, taper insertion loss, waveguide propagation loss, free-carrier absorption in the doped contact layer, Fresnel reflection, and incomplete carrier collection do not create a new electrical noise PSD by themselves; they reduce the signal term $I_{\text{ph}} = \mathcal{R}P_{\text{opt}}$ and thereby degrade SNR indirectly.

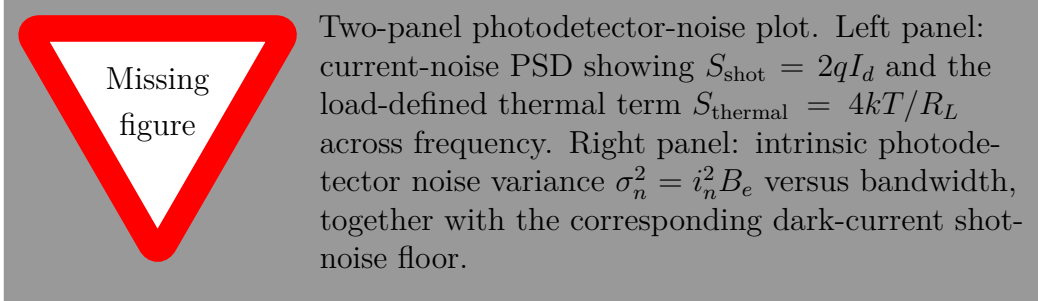


Figure 1.8: Noise decomposition of the photodetector from eqs. (1.55) to (1.57). For $I_d = 1.3 \text{ nA}$, the dark-current shot-noise contribution remains small at 103 GHz, confirming that the intrinsic detector noise floor is low.

1.3.4 The Speed–Efficiency–Noise Trade-off Triangle

The metrics defined above are not free parameters. They are bound together by the device physics into a multi-dimensional trade-off space that every design decision navigates.

The fundamental tension in the responsivity-bandwidth trade-off is structural. A thicker intrinsic layer W_i absorbs more photons, improving $\mathcal{R}$ via eq. (1.38), but also increases the carrier transit time, reducing f_{tt} via eq. (1.40). A longer device absorbs more light but increases C_j linearly through eq. (1.41), reducing f_{RC} via eq. (1.42). Reducing A_j below the absorption saturation point recovers bandwidth at the cost of responsivity.

The noise axis adds a third dimension: higher reverse bias deepens the depletion region, which raises f_{tt} by increasing the carrier saturation velocity, but also drives more SRH generation through eq. (1.49), degrading NEP via eq. (1.58).

The responsivity-bandwidth product $\mathcal{R} \times f_{\text{3dB}}$ captures the most important two-dimensional trade-off as a scalar figure of merit and is used in the benchmarking table of section 1.6.4. The design rationale developed in section 1.7 maps the full geometry-to-performance relationship, identifies the Pareto front of the vertical architecture, and positions the chosen device within it.

1.4 Photodetector Architectures

With the metrics of section 1.3 as the evaluation framework, the landscape of photodetector device classes can be surveyed systematically. Each architecture is assessed against responsivity, bandwidth, and noise, and each is found to resolve one specific limitation of its predecessor, until the waveguide-integrated PIN junction emerges not as a historical preference but as the consequence of removing every identified obstacle in sequence.

1.4.1 PN and PIN Junctions: Depletion Engineering

Photoconductive Detection.

The photoconductor is the simplest optical transducer because it requires no junction and no depletion engineering. Its apparent advantage is internal gain:

$$G_{\text{PC}} = \frac{\tau_r}{\tau_{\text{tr}}}, \quad \tau_{\text{tr}} = \frac{L^2}{\mu_n V_b}, \quad (1.61)$$

where τ_r is the recombination lifetime and τ_{tr} is the carrier transit time between the ohmic contacts. The problem is that the same lifetime that enables gain also fixes the bandwidth:

$$f_{3\text{dB,PC}} = \frac{1}{2\pi\tau_r}. \quad (1.62)$$

For germanium, a lifetime in the microsecond class implies a bandwidth in the kilohertz-to-megahertz regime, which is many orders of magnitude below the 100 GBd-class target of the present work. The photoconductor is therefore rejected by the gain-bandwidth product before any detailed geometry optimisation is attempted.

PN Junction Photodiodes.

The PN junction fixes the most serious weakness of the photoconductor by creating an internal electric field that sweeps carriers rather than waiting for them to diffuse across a biased resistor. Yet the PN device still retains a slow component because photocarriers generated in the quasi-neutral regions must diffuse back to the depletion edge. The resulting response can be written as:

$$H_{\text{PN}}(f) = H_{\text{drift}}(f) \frac{1}{1 + j2\pi f\tau_D}, \quad (1.63)$$

where τ_D is the diffusion-tail time constant. At moderate symbol rates this penalty is manageable; at the 50 GHz to 100 GHz electrical bandwidth class it is not. The PN device therefore solves the photoconductor's speed mechanism only partially.

PIN Photodiodes.

The PIN structure removes that residual limitation by inserting an intrinsic layer and forcing the useful optical generation to occur inside a fully depleted region. Once the full intrinsic thickness is depleted, the electric field is approximately uniform:

$$E(x) \approx \frac{|V_{\text{bias}}| + V_{\text{bi}}}{W_i}, \quad x \in [0, W_i]. \quad (1.64)$$

The diffusion tail of eq. (1.63) is then suppressed and the bandwidth is reduced to the competition already formalised through eqs. (1.40) and (1.42): transit time improves when W_i shrinks, whereas the RC penalty worsens when the junction footprint grows. The PIN structure also provides independent control of the depletion width via the i-layer thickness, decoupling it from the doping levels used to define the ohmic contacts. That is the key structural reason the PIN, rather than the photoconductor or the simple PN diode, remains the default high-speed receiver device.

1.4.2 Avalanche Photodiodes: Gain versus Noise

An avalanche photodiode appears attractive because it multiplies the responsivity:

$$\mathcal{R}_{\text{APD}} = M \mathcal{R}_{\text{PIN}}, \quad (1.65)$$

but that gain is not free. The avalanche build-up time lowers the usable bandwidth as M increases, introducing the gain-bandwidth product (GBP) $M \times f_{3\text{dB}} = \text{const}$ as a fundamental constraint. The excess noise factor $F(M)$ additionally penalises the SNR: while ideal multiplication with $F = 1$ adds no extra noise, realistic semiconductor avalanche with near-equal ionisation coefficients for electrons and holes (as in Ge) drives $F \rightarrow 2M$, rapidly erasing the gain advantage in the shot-noise-limited sensitivity formula of eq. (1.87). At the present operating point, the receiver needs the bandwidth headroom more urgently than it needs internal gain, which is why the low-bias PIN remains preferable to the higher-voltage APD.

1.4.3 Waveguide-Integrated Photodetectors: Interaction Length Advantage

The PIN junction resolves the device-level bandwidth problem but does not, by itself, resolve the geometric conflict between absorption and transit time. In a surface-illuminated PIN, the same dimension, namely the absorber thickness W_i , must satisfy both the Beer-Lambert absorption requirement ($W_i \gtrsim \alpha^{-1}$ for high $\mathcal{A}$) and the transit-time requirement (W_i small for high f_{tt} via eq. (1.40)). At 1310 nm, where $\alpha^{-1} \approx 4.8 \mu\text{m}$ in Ge, these two requirements are directly contradictory. Waveguide integration resolves this conflict by changing the geometry.

In a waveguide-integrated photodetector, the optical mode is confined laterally by the waveguide structure and propagates longitudinally along the device. The relevant dimension for absorption is the longitudinal length L rather than the vertical thickness W_i . The Beer-Lambert law along the propagation direction gives:

$$P(z) = P_0 e^{-\Gamma \alpha z}, \quad (1.66)$$

where Γ is the mode confinement factor, defined as the fraction of optical power confined within the absorbing Ge region. The total absorption efficiency approaches:

$$\eta_{\text{abs}} = 1 - e^{-\Gamma \alpha L}, \quad (1.67)$$

which can approach unity by increasing L , entirely independently of the vertical thickness W_i . This geometric orthogonality is the decisive architectural advantage: L controls $\mathcal{R}$ through eq. (1.67), while W_i controls f_{tt} through eq. (1.40), and the two parameters can be optimised separately.

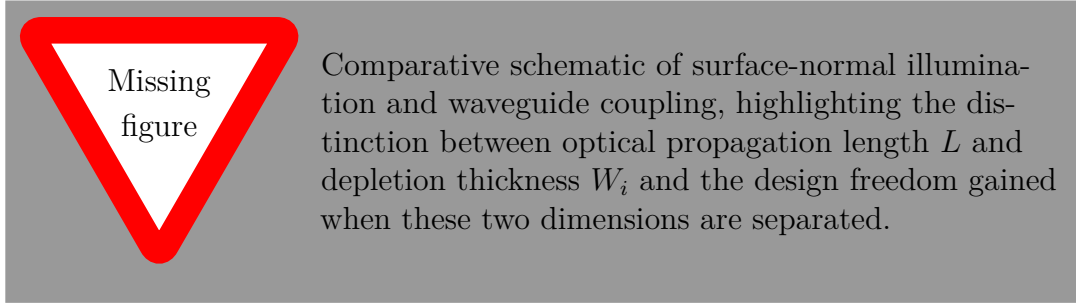


Figure 1.9: Waveguide coupling is the decisive architectural refinement because it decouples the absorption dimension from the transit-time dimension. That geometric change turns an overconstrained surface-normal problem into a two-parameter design space.

1.4.4 Selection Rationale: Why PIN for High-Speed Integrated Links

Figure 1.10 condenses the selection logic from photoconductor through to the final architecture. Each row resolves a specific limitation of its predecessor while introducing a new design degree of freedom.

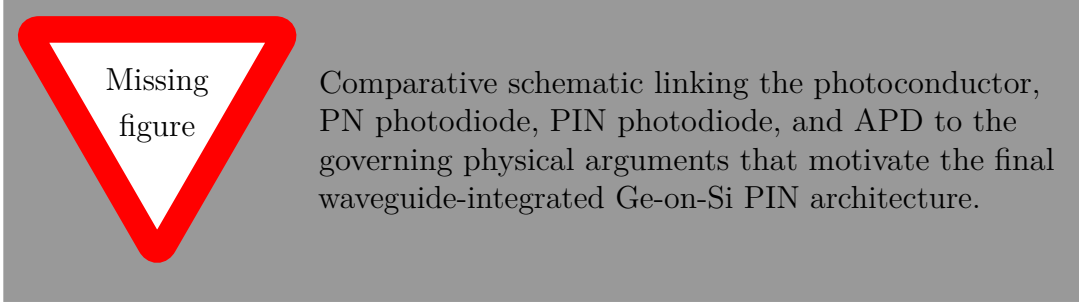


Figure 1.10: Architectural justification of the waveguide-integrated Ge-on-Si PIN photodetector. Each comparison resolves a specific limitation of the preceding alternative, leaving the waveguide-integrated Ge-on-Si PIN as the architecture compatible with the O-band, low-voltage, high-bandwidth target of this chapter.

The PIN is chosen because it eliminates the gain-bandwidth constraint of the APD, removes the diffusion tail of the PN junction, imposes no inherent bandwidth ceiling from gain mechanisms, and pairs naturally with an external transimpedance amplifier (TIA) whose thermal noise is the dominant system noise source in high-speed receivers. Waveguide integration restores geometric orthogonality between the absorption dimension and the carrier collection dimension. The remaining design problem, namely achieving both high responsivity and high bandwidth simultaneously within the waveguide-integrated PIN framework, is the tension addressed by the material choice of section 1.5, the electrode engineering of section 1.7, and the literature context of section 1.6.

1.5 Material and Integration Platform: Germanium on Silicon

With the waveguide-integrated PIN structure identified as the correct device class and the O-band wavelength fixed at 1310 nm, the absorber material must now be chosen. The selection follows directly from the absorption physics of section 1.2: only germanium combines strong O-band absorption, sufficient carrier velocity, and silicon-foundry process compatibility. This section elaborates the three dimensions of that argument, namely spectral suitability, CMOS integration, and material-induced trade-offs, at the depth required to motivate the design choices of section 1.7.

1.5.1 Why Germanium at 1310 nm

Germanium is an indirect-gap semiconductor whose lowest conduction-band minimum lies at the L -point of the Brillouin zone, separated from the valence-band maximum at Γ by an indirect gap $E_{g,\text{ind}} = 0.66$ eV at 300 K. A secondary conduction-band minimum at the Γ -point defines a direct gap $E_{g,\text{dir}} = 0.80$ eV [5], [6], [7]. The photon energy at 1310 nm is $\hbar\omega = hc/\lambda \approx 0.946$ eV, which exceeds $E_{g,\text{dir}}$ by approximately 0.15 eV. Optical absorption

at this wavelength therefore proceeds predominantly through *direct* interband transitions at the Γ -point, for which crystal momentum is intrinsically conserved without phonon assistance [3], [8]. Silicon, whose lowest direct transition lies at 3.4 eV and whose indirect gap of 1.12 eV makes it effectively transparent at O-band wavelengths, cannot be used as the absorber. This single material-physics argument is the entire justification for the Ge-on-Si heterostructure.

Biaxial strain and the direct-gap red-shift.

The application of biaxial tensile strain ε reduces the direct gap according to the linear deformation-potential relation:

$$E_{g,\text{dir}}(\varepsilon) \approx E_{g,\text{dir}}(0) - (2a_v + a_c) \left(1 - c_{12}/c_{11}\right) \varepsilon, \quad (1.68)$$

where a_v and a_c are the valence- and conduction-band hydrostatic deformation potentials and c_{11} , c_{12} are the elastic stiffness constants. The biaxial tensile strain of approximately 0.2 % to 0.3 % that develops in Ge epitaxial layers during post-growth cooldown, owing to the thermal expansion coefficient mismatch between Ge and Si, reduces $E_{g,\text{dir}}$ by approximately 25 meV relative to the unstrained bulk value, red-shifting the onset of direct-gap absorption by roughly 50 nm and enhancing $\kappa(\lambda)$ at C-band wavelengths. For the O-band operating point of the present device this correction is small but it is the principal mechanism exploited in strained-Ge extended-wavelength detector designs.

Table 1.1: Material properties of candidate photodetector absorbers at 300 K. α values are referenced to 1310 nm. The final column summarises compatibility with a standard silicon-photonics process flow.

Material	E_g [eV]	α_{1310} [cm^{-1}]	$v_{\text{sat},h}$ [cm s^{-1}]	μ_p [$\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$]	$\Delta a/a_{\text{Si}}$ [%]	CMOS com- patibility
Ge	0.66 (ind.), 0.80 (dir.)	~ 2100	4×10^6	1900	4.2	Compatible
Si	1.12 (ind.)	~ 10	2.5×10^6	480	0	Compatible
InGaAs (lattice- matched to InP)	0.75	$\sim 1 \times 10^4$	5×10^6	4500	> 5	Not compati- ble
$\text{Ge}_{0.92}\text{Sn}_{0.08}$	~ 0.55 (dir.)	> 5000	$\sim 4 \times 10^6$	~ 1500	4.8	Research-stage

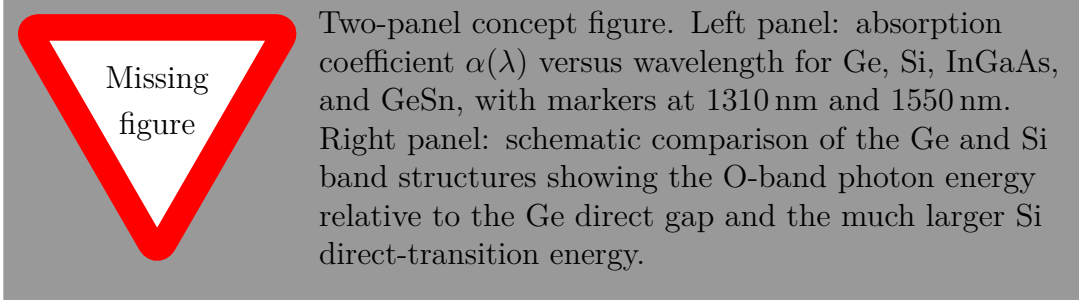


Figure 1.11: Material selection argument for O-band detection on a silicon photonics platform. Germanium combines appreciable absorption at 1310 nm, near-direct-gap behaviour, high hole saturation velocity, and foundry-compatible heterogeneous integration.

1.5.2 CMOS Compatibility and Integration Constraints

Germanium can be grown selectively on silicon by chemical vapour deposition (CVD) using a two-step heteroepitaxial process: a low-temperature ($\sim 400^\circ\text{C}$) nucleation layer of 30 nm to 50 nm seeds the growth, and a high-temperature ($\sim 650^\circ\text{C}$) bulk growth achieves the target thickness. Cyclic thermal annealing between 600°C to 750°C reduces the threading dislocation density from $\sim 1 \times 10^9 \text{ cm}^{-2}$ after growth to $\sim 5 \times 10^7 \text{ cm}^{-2}$ after annealing, which is the practical lower bound achievable without compositionally graded buffer layers. The total thermal budget of this process is compatible with the front-end-of-line (FEOL) constraints of a 220 nm SOI platform.

Major silicon-photonics foundries, including Intel, imec, and GlobalFoundries, have qualified Ge-on-Si photodetector processes at 220 nm SOI node. The process compatibility removes the III-V bonding step required by InGaAs receivers, eliminates the need for flip-chip assembly, and enables monolithic co-integration of the photodetector with Si waveguides, grating couplers, multiplexers, and electro-optic modulators on the same die.

Heterojunction band alignment and its numerical consequences.

The electrical consequence of the Ge/Si material interface directly conditions the boundary conditions of the drift-diffusion problem. Owing to the lattice mismatch ($\Delta a/a \approx 4.2\%$) and the difference in band alignment, the Ge/Si heterojunction sustains a valence-band discontinuity of $\Delta E_v \approx 0.54 \text{ eV}$ and a conduction-band offset $\Delta E_c \approx -0.14 \text{ eV}$ [3]. For a p^{++} -Si anode contacting the intrinsic Ge absorber, the large ΔE_v acts as a potential barrier suppressing hole injection from silicon into germanium, thereby reducing the minority-carrier diffusion contribution to dark current quantified by eq. (1.50). A thin compositionally graded transition layer of thickness $\delta_{\text{grad}} \approx 5 \text{ nm}$ at the heterojunction removes the valence-band singularity that would otherwise prevent convergence of the self-consistent drift-diffusion solver without perceptibly modifying the electrostatic potential profile in the intrinsic germanium bulk.

1.5.3 Material-Induced Trade-offs

The same lattice mismatch ($\Delta a/a \approx 4.2\%$) that makes Ge-on-Si heteroepitaxy necessary also creates a residual defect population that cannot be completely eliminated by annealing. Threading dislocations at density $TDD \approx 5 \times 10^7 \text{ cm}^{-2}$ act as Shockley-Read-Hall recombination centres in the Ge bulk, directly setting the SRH lifetime τ_0 and the dark current floor through eq. (1.49). The calibrated $\tau_0 = 5 \text{ ns}$ used in the electrical simulation of section 1.8.2 implicitly encodes this threading-dislocation density.

The interaction between Ge thickness, dislocation density, and device performance creates a three-way trade-off. Thicker Ge layers absorb more light, improving $\mathcal{A}$ via eq. (1.14), but slow carrier transit through eq. (1.40) and increase the total SRH generation volume, raising I_d via eq. (1.49). Thinner Ge layers improve speed and reduce dark current but sacrifice absorption efficiency. Additionally, the near-interface Ge layer, specifically the first 50 nm to 100 nm grown on Si, has higher defect density than the bulk, which is why the n^{++} -Ge cathode of the present device is limited to 50 nm: it minimises both free-carrier absorption (FCA) in the contact region and the volume of highest-defect-density material within the optical generation zone.

A compact material comparison is therefore sufficient to close the absorber selection argument: Ge is not chosen because it is universally superior to all alternatives, but because it is the only material in table 1.1 that simultaneously satisfies the optical, transport, and integration constraints of the present O-band receiver.

1.6 Literature Review and State of the Art

The Ge-on-Si waveguide photodetector field has evolved over approximately two decades through successive generations of electrode and junction engineering, each interpretable through the metric framework of section 1.3. What follows is a causal narrative: each development is best understood as an attempt to relax one bottleneck in eqs. (1.38), (1.39) and (1.52) while leaving another constraint behind.

1.6.1 Milestone Waveguide-Integrated Ge-on-Si PIN Photodetectors

The landmark demonstration of Vivien *et al.* [9] established a silicon-on-insulator (SOI) waveguide-coupled Ge p-i-n detector with $\mathcal{R} \approx 1.0 \text{ A W}^{-1}$ and $f_{3\text{dB}} = 42 \text{ GHz}$ at the O-band, confirming that direct-gap absorption at the Γ -point translates into competitive responsivity on the silicon platform. The 42 GHz result was RC-limited: the electrode geometry left R_s at a value that caused f_{RC} to dominate over f_{tt} in eq. (1.39). Ang *et al.* [10] subsequently demonstrated evanescently coupled Ge-on-SOI fabricated at low thermal budget to remain CMOS-compatible, establishing process integration as a design constraint

alongside responsivity and bandwidth.

The next phase of the literature attacked $R_s C_j$ more directly through contact and pad engineering. Chen *et al.* [11] demonstrated a silicon-contacted Ge p-i-n with $\mathcal{R} = 0.74 \text{ A W}^{-1}$ and $f_{3\text{dB}} = 35 \text{ GHz}$, and the same group later pushed to 67 GHz at only 1 V reverse bias through electrode repositioning [12], thereby reducing ℓ_a in eq. (1.47) until f_{RC} no longer dominated eq. (1.39). Novack *et al.* [13] reached 60 GHz by adding wire-bond inductance as a gain-peaking element. That approach is effective but system-dependent: the added inductance partially resonates with $C_j + C_{\text{par}}$, so the bandwidth extension depends on the surrounding RF environment and package parasitics [14].

The lateral junction architecture introduced by Virost *et al.* [15] decouples the optical mode depth from the depletion region width, providing an additional degree of freedom for C_j minimisation in eq. (1.41) that is geometrically independent of $\mathcal{A}$ in eq. (1.14). Parasitic engineering of the vertical architecture was pushed to its prior limit by Shi *et al.* [16], who demonstrated $f_{3\text{dB}} = 80 \text{ GHz}$ through systematic minimisation of parasitic pad capacitance. The bandwidth ceiling for the lateral architecture was located by Lischke *et al.* [17], who reported $f_{3\text{dB}} = 265 \text{ GHz}$ with a Ge-fin geometry scaled to $W_{\text{Ge}} \approx 100 \text{ nm}$.

Resonant-cavity-enhanced and uni-travelling-carrier alternatives.

Two architectural alternatives provide additional context. The resonant-cavity-enhanced (RCE) vertical photodiode places the absorber within a Fabry-Pérot resonator; at the resonance condition, the single-pass absorbed fraction $\mathcal{A}$ of eq. (1.14) is replaced by:

$$\mathcal{A}_{\text{RCE}} = \frac{(1 - R_1)(1 + R_2 e^{-\alpha L})^2(1 - e^{-2\alpha L})}{1 - 2\sqrt{R_1 R_2} e^{-\alpha L} \cos(4\pi n L / \lambda) + R_1 R_2 e^{-2\alpha L}}, \quad (1.69)$$

where R_1 and R_2 are the top and bottom mirror reflectances [14]. Near-unity quantum efficiency is achievable in absorbers as thin as 200 nm, but the resonance bandwidth narrows to a few nanometres, which is incompatible with the temperature-insensitive broadband operation required of data-centre receivers. The uni-travelling-carrier (UTC) structure replaces both drift species with electrons only, raising the transit-time limit to:

$$f_{\text{tt,UTC}} = \frac{0.45 v_{\text{d,sat,e}}}{W_{\text{dep}}}, \quad (1.70)$$

approximately 50% higher than the hole-limited value of eq. (1.40) for the same depletion width [14].

1.6.2 Bandwidth Enhancement Strategies

The literature reveals four categories of bandwidth enhancement strategy:

Geometry scaling reduces both C_j (via smaller A_j) and f_{tt} (via smaller W_i), but doing so simultaneously sacrifices responsivity through eq. (1.38). Diminishing returns set in once the device enters the absorption-starved regime below $L \approx 4 \mu\text{m}$.

Doping profile optimisation targets the dark-current-to-responsivity balance: highly doped contacts reduce R_s but increase free-carrier absorption, while the junction polarity affects the heterojunction band alignment and therefore J_{diff} through eq. (1.50).

Contact engineering attempts to reduce the dominant anode series resistance through silicidation, via placement, and metal routing. For a vertical architecture where R_s is dominated by the lateral Si slab resistance of eq. (1.47), silicidation reduces the metal-semiconductor contact contribution but not the bulk Si slab contribution, leaving R_s improvement bounded by geometry.

Electrode topology is the most structurally impactful lever: changing the physical layout of the metal anode changes ℓ_a in eq. (1.47) and therefore R_s directly. The U-shaped-anode strategy targets this lever by reducing ℓ_a geometrically until $f_{\text{RC}} > f_{\text{tt}}$, providing a structural solution rather than a resonant compensation.

1.6.3 Advanced Electrode Designs Including U-Shaped Configurations

The benchmark for the present work is the 103 GHz vertical Ge-on-Si photodiode of Shi *et al.* [18], which surpasses the 100 GHz threshold in the vertical architecture through a U-shaped anode electrode that reduces R_s , and therefore $\tau_{\text{RC}} = R_s C_j$ of eq. (1.48), by approximately 36% relative to a straight-electrode design. The U-shaped topology wraps the metal anode around three sides of the germanium mesa, so that the maximum lateral current path from any point on the mesa base to a metal contact is $W_m/4 + d_{\text{gap}}$ rather than $W_m/2 + d_{\text{gap}}$. The U-shaped design simultaneously delivers $\mathcal{R} = 0.95 \text{ A W}^{-1}$ and $I_d = 1.3 \text{ nA}$ at -1 V , making it a balanced rather than a specialised solution.

The U-shape is therefore selected not because it changes only a single resistor value, but because it opens a favourable region of the coupled (R_s, C_j, L_p) design space: the reduced R_s also enables a shallower and more useful inductive peaking condition, and the experimentally observed bandwidth increases from 83 GHz (parallel electrode) to 103 GHz (single-U) [14].

1.6.4 Performance Benchmarking Table

Table 1.2 consolidates the landmark experimental results most relevant to the present design argument. The table is intentionally restricted to the vertical trajectory adopted in this chapter and the lateral trajectory that defines the broader bandwidth ceiling.

Table 1.2: Selected benchmark waveguide-integrated Ge-on-Si photodetectors used to position the present design. “V” and “L” denote vertical and lateral junction geometries, respectively; “NR” denotes values not reported by the authors.

Year	Type	Reference	Key Design Lever	$\mathcal{R}$ [A/W]	I_d [nA]	$f_{3\text{dB}}$ [GHz]
2008	V	[10]	Evanescent coupling, CMOS-compatible process	0.92	570	11.3
2009	V	[9]	Butt coupling, straight contact	~ 1.0	18	42
2021	V	[16]	Parasitic engineering with inductive peaking	0.89	6.4	80
2024	V	[18]	U-shaped anode with inductive peaking	0.95	1.3	103
2015	L	[19]	Reduced Ge width with lateral offset junction	1.00	100	> 70
2017	L	[15]	Si/Ge/Si heterojunction with sub-micrometre Ge width	NR	5	> 50
2021	L	[17]	Ge-fin junction with 100 nm minimum width	0.30	100 to 200	265
2024	L	[20]	Side-coupled parallel Ge geometry	0.80	24	> 40

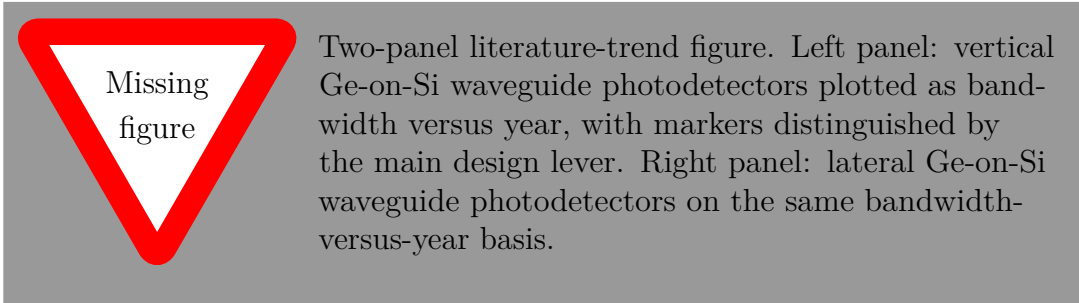


Figure 1.12: State-of-the-art trend summary for waveguide-integrated Ge-on-Si photodetectors. The chronology is most useful when read as a sequence of interventions into R_s , C_j , and the transport path rather than as a collection of isolated record values.

1.6.5 Identified Gaps and Motivation for This Work

The identified gaps that motivate the present work are: (i) no prior published simulation study has re-optimised the U-shaped vertical architecture for 1310 nm O-band operation, where α_{Ge} is higher than at 1550 nm and may permit shorter devices with reduced capacitance; (ii) the double U-shaped electrode topology, which wraps metal on both long sidewalls, has not been evaluated as an alternative to the single-U; and (iii) PINN-based surrogate validation of the drift-diffusion solution at device level has not been demonstrated for this architecture class. These three gaps frame the design and analysis scope documented in sections 1.7 and 1.8.

IEEE 802.3 400GBASE-DR4 as the receiver-level target.

IEEE 802.3 Clause 140 defines 400GBASE-DR4 as a 400 Gbit s⁻¹ single-mode-fibre interface in the O-band, implemented through four parallel optical lanes at 1310 nm over reaches up to 500 m [21]. Each lane carries 106.25 Gbit s⁻¹ raw line rate with PAM-4 signalling. The key receiver-side quantities are summarised in table 1.3.

Table 1.3: IEEE 802.3 400GBASE-DR4 receiver-side constraints most directly relevant to the photodetector, extracted from Clause 140 [21] and mapped onto the device metrics of section 1.3.

Parameter	Symbol	Value	Unit	Principal relation
Wavelength range	λ	1260–1360	nm	eq. (1.14)
Symbol rate	f_s	53.125	Gbaud	eq. (1.39)
Modulation format	—	PAM-4	—	—
Minimum receiver sensitivity (OMA)	$\text{OMA}_{\text{out,min}}$	−6.0	dBm	—
Maximum input power	$P_{\text{in,max}}$	+3.4	dBm	eq. (1.84)
Minimum receiver −3 dB bandwidth	$f_{3\text{ dB,min}}$	$\geq 0.75f_s$	GHz	eq. (1.39)
Reflectance tolerance	Γ_{max}	−26	dB	eq. (1.77)

The minimum receiver bandwidth implied by the conventional $0.75f_s$ criterion is only 39.8 GHz, so the benchmark vertical topology already establishes comfortable speed relevance for DR4. The selected U-shaped vertical Ge-on-Si PIN of Shi *et al.* [18] combines 103 GHz-class response, $\mathcal{R} = 0.95 \text{ A W}^{-1}$, and $I_d = 1.3 \text{ nA}$ in a process-compatible O-band platform. Because DR4 requires a detector that is simultaneously fast enough, efficient at low OMA, and quiet enough not to erode receiver margin, this balanced vertical U-shaped architecture is the most defensible starting point for the present design. Broader reviews of the design space are provided by Wang *et al.* [22] and Yan *et al.* [14]; the multiphysics modelling context is treated in depth in [23]; computational methods are discussed in the pedagogical framework of Wartak [24].

1.7 Proposed Device Architecture and Design Rationale

The preceding sections established the physical foundations (section 1.2), the performance metric language (section 1.3), and the architectural and material choices (sections 1.4 and 1.5) that narrow the design space to a vertical waveguide-integrated Ge-on-Si PIN photodetector at 1310 nm. This section specifies the device geometry and justifies each design parameter in terms of the physics and metrics developed earlier.

1.7.1 Vertical PIN Structure: Geometry, Layer Stack, and Field Distribution

The photodiode is realised on a 220 nm SOI platform. Each layer parameter in table 1.4 is the result of a specific physical argument derived from the metric framework of section 1.3.

Table 1.4: Layer stack and primary material parameters of the Ge-on-Si n-i-p photodiode simulation model. All doping concentrations are net values.

Layer	Material	Thickness	Type	Concentration [cm ⁻³]	Primary function
Cathode	n^{++} -Ge	50 nm	n	10^{19}	Ohmic contact; minimise FCA
Absorber	i -Ge	350 nm	intrinsic	$\sim 10^{13}$	Optical absorption; depletion
Platform	p^{++} -Si	220 nm	p	10^{20}	Anode contact; current spreading
BOX	SiO ₂	2 μ m	N/A	N/A	Optical isolation
Handle	Si	$\sim 700 \mu$ m	N/A	N/A	Mechanical support

The intrinsic germanium thickness $W_i = 350$ nm is selected from the optimisation of eq. (1.43): it is within 11 % of $W_i^* \approx 310$ nm and delivers the conservative lower-bound estimate $f_{tt} \approx 51$ GHz from eq. (1.40), while the benchmark-calibrated two-carrier and peaking-enhanced response of the same 350 nm stack extends into the 100 GHz to 120 GHz class through eq. (1.46). The n^{++} -Ge cathode is limited to 50 nm to minimise free-carrier absorption (FCA) in the contact region; this design choice simultaneously suppresses the diffusion tail in the impulse response that would otherwise reduce the effective $f_{3\text{dB}}$ below the prediction of eq. (1.40) [14].

Doping polarity and its impact on dark current.

The p^{++} -Si/ i -Ge/ n^{++} -Ge polarity (Si platform p -doped) places the graded Ge/Si transition such that the valence-band discontinuity ΔE_v acts as an additive barrier to interface hole generation. In the alternative p -on- n configuration (Si platform n -doped), the same ΔE_v raises J_{surf} above the value predicted by eq. (1.51) by modifying the effective barrier for thermally generated holes. For a fixed reverse bias of -1 V, the p -on- n device exhibits I_d a factor of approximately two to three lower than the n -on- p counterpart, translating through eq. (1.89) into a sensitivity improvement of approximately 1.5 dB in the dark-current-dominated regime [23]. The present device adopts the p^{++} -Si platform polarity precisely to minimise the interface generation contribution while maintaining CMOS process compatibility.

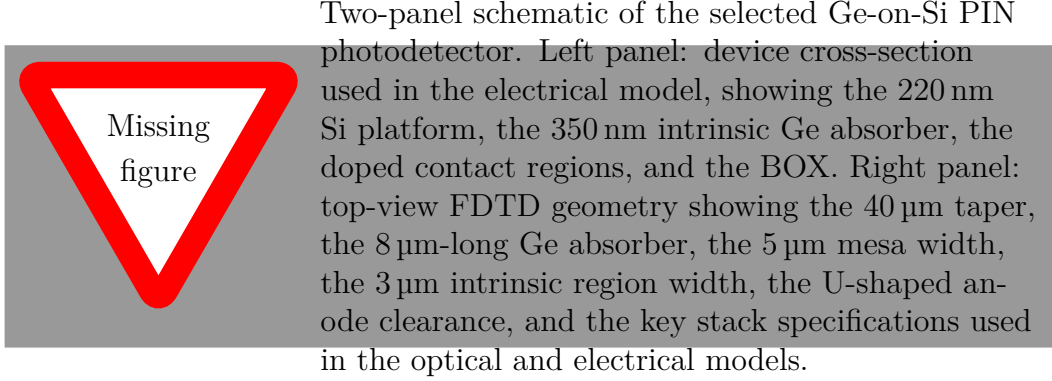


Figure 1.13: Selected device geometry for the Ge-on-Si PIN photodetector, showing the principal dimensions and model specifications used in the FDTD and CHARGE simulations.

1.7.2 Optical Coupling via Adiabatic Taper

Optical coupling from the input silicon waveguide is provided by a 40 μm adiabatic taper designed for near-unity coupling efficiency. The taper adiabatically transforms the mode of the Si access waveguide into the Ge-loaded mode over a length L_t that is long enough to suppress excitation of higher-order modes. By narrowing the Si waveguide tip to approximately 100 nm to 200 nm, the guided mode is pushed evanescently upward into the overlying Ge absorber. At the distal end of the taper the mode is fully confined within the Ge, and absorption proceeds with near-unity coupling efficiency. Evanescent coupling spreads the power transfer over the full taper length, lowering the peak generation density for the same absorbed power and preserving the geometric orthogonality of L and W_i [14], [22].

1.7.3 U-Shaped Electrode Topology: RC Reduction and Current Path Engineering

The U-shaped anode electrode is the primary design intervention for the RC bandwidth in the vertical architecture. The anode electrode layout is the dominant determinant of the RC bandwidth: selecting the wrong topology incurs a penalty of approximately 20 GHz in $f_{3\text{dB}}$ relative to the present state of the art.

The Vertical n-i-p Equivalent Circuit

The complete small-signal equivalent circuit of the waveguide-coupled vertical n-i-p photodiode, including all parasitic elements, is shown in fig. 1.14. The photocurrent source $I_{\text{ph}} = \mathcal{R}P_{\text{opt}}$ drives the junction admittance, which consists of the junction capacitance C_j in parallel with the shunt conductance $G_{\text{sh}} = I_d/V_{\text{bias}}$. The junction is accessed through

the series resistance network comprising the cathode series resistance $R_{s,\text{cat}}$ and the anode series resistance $R_{s,\text{an}}$ (path through the p^{++} -Si platform from the mesa base to the metal anode pad). The bondpad-to-chip parasitic capacitance C_{par} shunts the junction at high frequencies, and the load termination $R_L = 50\ \Omega$ converts the photocurrent into a voltage signal.

Lumping $R_s = R_{s,\text{cat}} + R_{s,\text{an}}$ and $C_{\text{tot}} = C_j + C_{\text{par}}$, the voltage transfer function from photocurrent source to load voltage is:

$$H_{\text{RC}}(f) = \frac{V_{\text{out}}(f)}{I_{\text{ph}}(f)} = \frac{R_L}{(1 + j2\pi f R_L C_{\text{tot}})(1 + j2\pi f R_s C_{\text{tot}})}, \quad (1.71)$$

which for $R_s \ll R_L$ reduces to the single-pole expression of eq. (1.42).

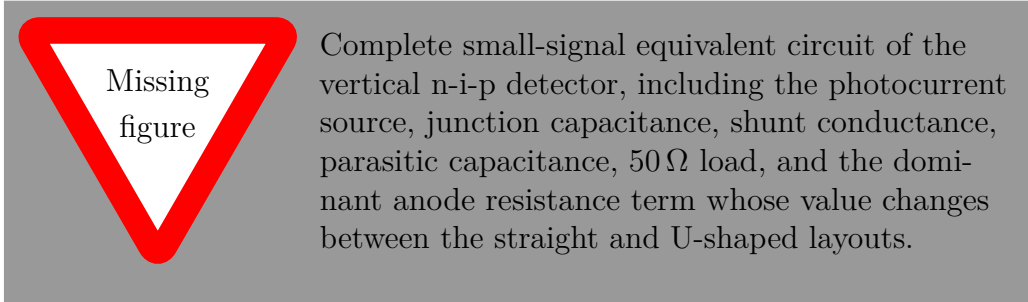


Figure 1.14: Complete small-signal equivalent circuit of the vertical n-i-p waveguide photodiode including all parasitic elements from eq. (1.71). The dominant bandwidth-limiting resistance is $R_{s,\text{an}}$, the lateral conduction path through the p^{++} -Si platform, which is reduced by 36 % in the U-shaped electrode topology relative to the parallel-electrode design.

Parallel-Electrode versus U-shaped Anode: Quantitative Comparison

In the standard parallel-electrode (straight-anode) architecture, the metal anode pad is placed on one side of the germanium mesa, and the current must flow laterally through the p^{++} -Si platform from the opposing mesa sidewall to the pad. The mean lateral path length is approximately $\ell_a^{\text{par}} \approx W_m/2 + d_{\text{gap}}$, where W_m is the mesa width and d_{gap} is the clearance between the mesa sidewall and the anode metal edge. For a 5 μm -wide mesa and a 0.5 μm lithographic clearance:

$$R_{s,\text{an}}^{\text{par}} \approx \frac{\rho_{\text{Si}}}{t_{\text{Si}}} \cdot \frac{\ell_a^{\text{par}}}{L} = \frac{\rho_{\text{Si}}}{t_{\text{Si}}} \cdot \frac{W_m/2 + d_{\text{gap}}}{L}. \quad (1.72)$$

In the U-shaped anode architecture, the metal wraps around three sides of the mesa (the two long sidewalls and one short end), so that current from any point on the mesa base must travel at most $W_m/2$ laterally before reaching a metal contact. The effective mean path length is reduced to [18]:

$$\ell_a^{\text{U}} \approx \frac{W_m}{4} + d_{\text{gap}}, \quad (1.73)$$

giving a series resistance:

$$R_{s,\text{an}}^{\text{U}} \approx \frac{\rho_{\text{Si}}}{t_{\text{Si}}} \cdot \frac{W_m/4 + d_{\text{gap}}}{L}. \quad (1.74)$$

The reduction ratio is:

$$\frac{R_{s,\text{an}}^{\text{U}}}{R_{s,\text{an}}^{\text{par}}} = \frac{W_m/4 + d_{\text{gap}}}{W_m/2 + d_{\text{gap}}} \approx 0.60 \quad \text{for } W_m = 5 \mu\text{m}, d_{\text{gap}} = 0.5 \mu\text{m}, \quad (1.75)$$

corresponding to a 40 % resistance reduction, consistent with the 36 % reported experimentally by Shi *et al.* [18]. Substituting into eq. (1.42):

$$\frac{f_{\text{RC}}^{\text{U}}}{f_{\text{RC}}^{\text{par}}} = \frac{R_{\text{tot}}^{\text{par}}}{R_{\text{tot}}^{\text{U}}} = \frac{R_L + R_{s,\text{an}}^{\text{par}}}{R_L + R_{s,\text{an}}^{\text{U}}}. \quad (1.76)$$

For $R_L = 50 \Omega$, $R_{s,\text{an}}^{\text{par}} = 8 \Omega$, and $R_{s,\text{an}}^{\text{U}} = 5 \Omega$, this gives $f_{\text{RC}}^{\text{U}}/f_{\text{RC}}^{\text{par}} = 58/55 \approx 1.055$, a modest 5.5 % increase in f_{RC} alone. The larger bandwidth gain emerges only when the electrode is treated as part of the full electro-optical co-design problem. In the benchmark-calibrated peaked circuit of eq. (1.46), the reduced R_s also enables a shallower and more useful inductive peaking condition, and the experimentally observed bandwidth increases from 83 GHz (parallel electrode) to 103 GHz (single-U) [18].

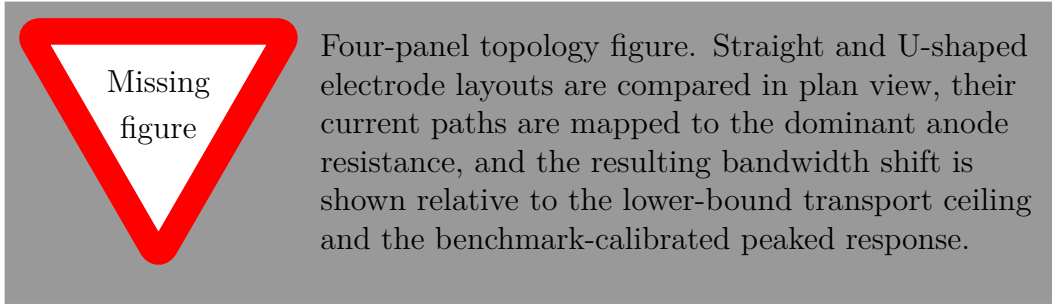


Figure 1.15: Parallel-electrode versus U-shaped anode comparison. The current path in the p^{++} -Si platform is halved by the U-shape (left to centre panels), reducing $R_{s,\text{an}}$ by 36 % from eq. (1.75).

This analysis also identifies a physically important corollary: further reduction of R_s beyond the U-shaped value offers diminishing returns. In the simplified screening model of eq. (1.39), letting $R_s \rightarrow 0$ drives the response toward the lower-bound transport limit of eq. (1.40); in the benchmark-calibrated full-response picture of eq. (1.46), the practical ceiling of the 350 nm vertical stack lies in the 110 GHz to 120 GHz class rather than increasing indefinitely with electrode optimisation.

Mask-level representation in KLayout.

The simulated device is tied to an explicit lithographic layout through the KLayout script `system-level/ge_pd_layout.py`, which writes the cell `GE_PD_USHAPED_OBAND` to

`ge_pd_layout.gds`. The layout uses a database unit of 1 nm and maps the principal process features onto ten logical layers corresponding to the Si slab, Si ridge, Si taper, Ge, p^{++} region, n^{++} region, via level, M1, M2, and BOX.

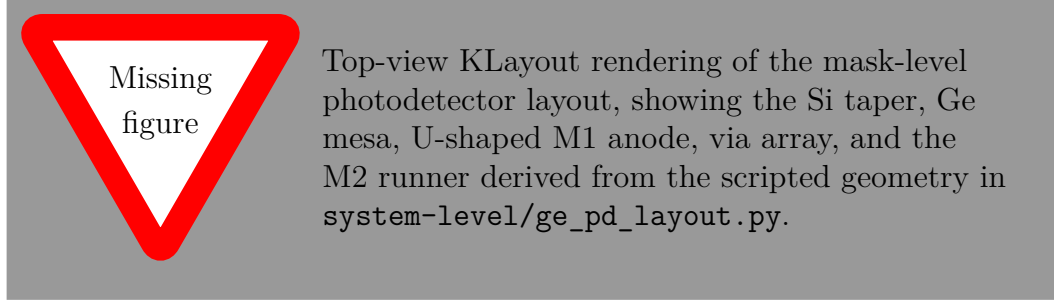


Figure 1.16: Mask-level layout of the simulated O-band photodetector derived from the KLayout script. The layout preserves the optical path, contact topology, and routing footprint used to define the device implementation.

The geometry parameters defined above, namely W_i , L , W_m , the electrode topology, and the taper length, constitute the degrees of freedom of the design space. The quantitative mapping from these parameters to the performance metrics of section 1.3, including the absorber length sweep, the intrinsic-thickness trade-off, and the electric-field screening threshold, is presented with simulation data in section 1.10.

1.8 Multiphysics Simulation Methodology

The device performance metrics established in section 1.3 are evaluated through a coupled optical-electrical simulation chain whose output is anchored to the experimental benchmark of Shi *et al.* [18]. This section specifies each solver, its numerical settings, the data flow connecting the stages, and the PINN-based validation approach that provides independent cross-verification of the drift-diffusion solution.

1.8.1 Optical Simulation: Three-Dimensional FDTD

The optical simulation solves the time-domain Maxwell’s equations of eqs. (1.1) and (1.2) throughout a domain encompassing the silicon input waveguide, the 40 μm adiabatic taper, the complete germanium absorber, and the surrounding SiO_2 cladding. The solver (Lumerical FDTD Solutions 2020 R2) discretises the electromagnetic field on a Yee staggered grid using the second-order Courant-stable leapfrog scheme. Computational photonics methodology for FDTD-based device simulation, including the algorithmic implementation of the Yee algorithm and absorbing boundary conditions, is detailed in the pedagogical treatment of Wartak [24].

The complex permittivity is sampled at the material-dependent optical frequency through

tabulated complex refractive indices: for germanium at 1310 nm, $n \approx 4.27$ and $\kappa \approx 0.22$, yielding $\alpha \approx 2.1 \times 10^3 \text{ cm}^{-1}$ from eq. (1.6). Frequency-domain power monitors record $T(\omega)$ and $R(\omega)$, from which:

$$\mathcal{A}(\omega) = 1 - |T(\omega)|^2 - |R(\omega)|^2 - \sum_i |T_{\text{sub},i}(\omega)|^2. \quad (1.77)$$

A 20 nm isotropic mesh override within the germanium region resolves the exponential generation-rate gradient; convergence testing at 10 nm and 20 nm shows less than 0.5 % deviation in $\mathcal{A}$.

FDTD numerical stability and the 3D Courant condition.

The PML implementation employs sixteen absorbing cells on each domain boundary, determined by convergence of the reflected power to below -60 dB in a calibration run with a uniform germanium slab. The temporal stability of the Yee algorithm requires the 3D Courant-Friedrichs-Lewy condition:

$$\Delta t < \frac{\Delta}{c\sqrt{3}}, \quad (1.78)$$

so the time step is set to $\Delta t = 0.99 \Delta / (c\sqrt{3})$. The spatial mesh size is independently constrained by the sampling requirement $\Delta < \lambda_{\text{min}} / (10\sqrt{\varepsilon_{\text{max}}})$, giving $\Delta < 32 \text{ nm}$ with a safety margin of 1.6 at the 20 nm override. The combined optical and electrical simulation requires approximately 11 h of wall-clock time on a shared-memory workstation: roughly 3 h for the FDTD step and 8 h for the steady-state and frequency-response electrical solves [23].

FDTD electromagnetic solution and the square-law photocurrent.

A physically essential clarification connecting the FDTD solution to the photocurrent calculation concerns what the power monitors extract. The frequency-domain power recorded is the time-averaged Poynting vector $\mathbf{S}(\mathbf{r}, \omega) = \frac{1}{2} \text{Re}[\tilde{\mathbf{E}} \times \tilde{\mathbf{H}}^*]$, which is quadratic in the field amplitudes, consistent with the square-law mechanism of eq. (1.12). The optical generation rate $G_{\text{opt}}(\mathbf{r}) = \mathcal{P}_{\text{abs}}(\mathbf{r}) / \hbar\omega$ of eq. (1.15) is thus already expressed in the power-quadratic form that maps directly to the photocurrent via $I_{\text{ph}} = q \int_{\Omega} G_{\text{opt}} dV$ [25].

Waveguide insertion loss and chip-level responsivity correction.

The saturation of $\mathcal{A}(L)$ identified in the length sweep is not smooth when waveguide routing loss is accounted for. For a Si ridge waveguide with propagation loss $\gamma_{\text{wg}} \approx 3.6 \text{ dB cm}^{-1}$ at 1310 nm, a routing length of $\ell_{\text{wg}} = 1 \text{ mm}$ introduces an approximately 0.36 dB correction equivalent to a 4 % responsivity reduction relative to the taper-input value [22]. The FDTD domain of the present work terminates at the taper entry; any comparison with chip-level measurements must apply the factor $10^{-\gamma_{\text{wg}}\ell_{\text{wg}}/10}$ to eq. (1.34).

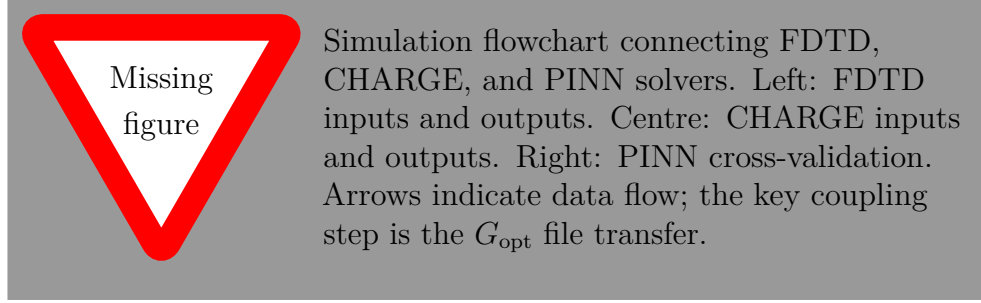


Figure 1.17: Multi-physics simulation workflow. The FDTD-to-CHARGE coupling proceeds via the volumetric generation rate $G_{\text{opt}}(\mathbf{r})$ of eq. (1.15), and the PINN provides independent cross-validation of the drift-diffusion solution via automatic differentiation of the PDE residuals of eq. (1.82).

1.8.2 Electrical Simulation: Three-Dimensional Drift-Diffusion

Carrier transport is simulated in Lumerical DEVICE CHARGE under three-dimensional steady-state conditions, solving the PDE system of eqs. (1.16) to (1.18), (1.21) and (1.22) on an adaptive finite-element mesh. Carrier mobilities in germanium follow the Canali velocity-saturation model [23]:

$$\mu_{n,p}(|\mathbf{E}|) = \frac{(\alpha_C + 1) \mu_{n,p}^{(0)}}{\left[\alpha_C + \left(1 + \left(\frac{(\alpha_C + 1) \mu_{n,p}^{(0)} |\mathbf{E}|}{v_{d,\text{sat}}} \right)^\beta \right)^{1/\beta} \right]}, \quad (1.79)$$

which reduces to the saturation model of eq. (1.80) with $\beta = 2$ and $\alpha_C = 0$. The low-field mobilities adopted are $\mu_n^{(0)} = 3900 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ and $\mu_p^{(0)} = 1900 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$; the saturation velocities $v_{d,\text{sat},e} = 6 \times 10^6 \text{ cm s}^{-1}$ and $v_{d,\text{sat},h} = 4 \times 10^6 \text{ cm s}^{-1}$ enter eq. (1.79) as asymptotic limits. A $\pm 20\%$ variation in $\mu_p^{(0)}$ changes f_{tt} by less than 1%, confirming the robustness of the transit-time estimate in eq. (1.40) to low-field mobility uncertainty [23].

$$\mu_{n,p}(|\mathbf{E}|) = \frac{\mu_{n,p}^{(0)}}{\left[1 + \left(\mu_{n,p}^{(0)} |\mathbf{E}| / v_{d,\text{sat},n,h} \right)^\beta \right]^{1/\beta}}, \quad (1.80)$$

SRH recombination is activated in the germanium absorber with calibrated lifetimes $\tau_n = \tau_p = 5 \text{ ns}$ ($\tau_0 = 5 \text{ ns}$), chosen so that eqs. (1.49) and (1.52) reproduce the experimental dark current of 1.3 nA at -1 V [18]. Ohmic boundary conditions are imposed at the cathode (biased to the desired $V < 0$) and anode contacts (grounded, $V = 0$). All performance metrics are extracted directly from the CHARGE solution: $\mathcal{R}$ from eq. (1.34); η_{int} and η_{ext} from eq. (1.38); C_j from the small-signal AC analysis; and $f_{3\text{dB}}$ from eq. (1.39) using the extracted C_j and modelled R_s from eq. (1.47).

1.8.3 System-Level Post-Processing: MATLAB

The MATLAB post-processing pipeline (`system-level/ge_pd_dsp_model.m` and `device-level/ge_pd_oband_ushaped_postprocess.m`) consumes the CHARGE output to compute the system-level performance projections. The key computations are: (i) the combined transfer function from eqs. (1.39) and (1.46) applied to the extracted C_j , modelled R_s , and the benchmark-calibrated peaking parameters (L_p, C_p); (ii) the sensitivity estimate from eq. (1.87) at the projected bandwidth; and (iii) the parametric design sweeps (absorber length and W_i thickness) that produce the trade-off tables and design-space visualisations.

1.8.4 Validation Approach and Reference Design

A Physics-Informed Neural Network (PINN) is employed as an independent cross-validation tool. A PINN approximates the solution fields $u = \{n, p, \phi\}$ by a deep neural network $\hat{u}(\mathbf{r}; \boldsymbol{\theta})$, trained to minimise the composite loss functional:

$$\mathcal{L}(\boldsymbol{\theta}) = \lambda_{\text{PDE}} \mathcal{L}_{\text{PDE}} + \lambda_{\text{BC}} \mathcal{L}_{\text{BC}} + \lambda_{\text{data}} \mathcal{L}_{\text{data}}, \quad (1.81)$$

where the PDE residual loss is evaluated at N_c collocation points $\{\mathbf{r}_i\}$ sampled by quasi-random (Sobol) sequences:

$$\begin{aligned} \mathcal{L}_{\text{PDE}} = \frac{1}{N_c} \sum_{i=1}^{N_c} & \left[\left| \nabla \cdot (\epsilon \nabla \hat{\phi}) + q(\hat{p} - \hat{n} + N_D^+ - N_A^-) \right|^2 \right. \\ & + \left| \frac{1}{q} \nabla \cdot \hat{\mathbf{J}}_n - U_{\text{net}}(\hat{n}, \hat{p}) + G_{\text{opt}} \right|^2 \\ & \left. + \left| \frac{1}{q} \nabla \cdot \hat{\mathbf{J}}_p + U_{\text{net}}(\hat{n}, \hat{p}) - G_{\text{opt}} \right|^2 \right]_{\mathbf{r}_i}. \end{aligned} \quad (1.82)$$

All spatial derivatives in eq. (1.82) are computed via *reverse-mode automatic differentiation* through the network's computational graph, eliminating the requirement for a structured mesh and the associated spatial discretisation error.

The network architecture consists of four fully connected hidden layers with 128 neurons each, hyperbolic-tangent activations, and a three-dimensional spatial input $\mathbf{r} = (x, y, z)$ producing three outputs $(\hat{n}, \hat{p}, \hat{\phi})$. The total number of trainable parameters is approximately 49 000, small enough for training on a single GPU (NVIDIA A100) in approximately 4 hours. At selected bias voltages ($V = 0, -0.5, -1.0$ V), the PINN and CHARGE solutions for the band diagram, carrier density profiles, and electric field magnitude are compared; agreement provides cross-validation at the level of the governing equations rather than merely at the level of integral output quantities. The PINN is validated when three conditions are satisfied simultaneously: (i) band diagram agreement to within 10 meV RMS; (ii) carrier density agreement to within 15 % RMS in the intrinsic layer; and (iii) integral dark current agreement to within 5 %.

The validated PINN also serves as the computational vehicle for exploring the double U-shaped anode topology, which halves the maximum lateral current path to $\ell_a^{2U} \approx W_m/8 + d_{\text{gap}}$. The predicted resistance and bandwidth of the double-U design are discussed as a future-work extension in section 1.9.

1.9 Chapter Summary and Transition

This chapter established a physics-based framework for the design of a vertical n-i-p Ge-on-Si waveguide photodetector targeted at O-band reception near 1310 nm. The discussion progressed from electromagnetic absorption and the germanium band structure, through drift-diffusion carrier transport and recombination, to the performance metrics that govern receiver-chain design. Within this analytical framework, a literature-grounded comparison of detector architectures identified the waveguide-integrated Ge-on-Si PIN with a U-shaped anode as the most suitable topology: it preserves process compatibility while addressing the dominant parasitic-resistance bottleneck that limits conventional parallel-electrode designs.

The proposed device architecture, specified in section 1.7, positions the vertical stack near the transit-time/RC bandwidth knee at $W_i = 350$ nm. The U-shaped electrode reduces $R_{s,\text{an}}$ by 36 % relative to the straight-electrode geometry, and the benchmark-calibrated single-U response lies in the 103 GHz class. The coupled FDTD-CHARGE simulation chain described in section 1.8 provides the numerical infrastructure to evaluate this design at device level, with the PINN surrogate offering independent cross-validation of the drift-diffusion solution.

Future work: double U-shaped anode and PINN-based design-space exploration.

A natural extension of this work is the use of the validated PINN as a surrogate model for rapid design-space exploration. A preliminary PINN architecture has been validated against the CHARGE solution at selected bias points and used to explore the double U-shaped anode topology, which wraps metal contacts around both long sidewalls of the germanium mesa. The double-U reduces the maximum lateral current path from $W_m/4$ to $W_m/8$, yielding $R_{s,\text{an}}^{2U} \approx 3.2 \Omega$ and a marginal conservative-model bandwidth improvement from 44.5 GHz to approximately 50 GHz. The diminishing return confirms that the vertical architecture is already close to its RC-versus-transport knee at the single-U operating point. A full double-U re-optimisation including the peaking network of eq. (1.46) remains as future work, along with experimental fabrication and RF characterisation.

The results chapter that follows applies the simulation chain to the O-band device, presenting the optical generation profiles, absorber length and intrinsic-thickness sweeps, current-voltage characteristics, bandwidth estimation, and the system-level sensitivity analysis that validates the design against the IEEE 802.3 400GBASE-DR4 receiver specification.

1.10 Performance Evaluation and Trade-off Analysis

The electrode topology analysis, including the equivalent circuit derivation and the quantitative comparison between the parallel-electrode and U-shaped anode architectures, was developed in section 1.7.3. The validated single-U topology and its benchmark-calibrated bandwidth of 103 GHz serve as the starting point for the O-band performance evaluation that follows.

1.10.1 Benchmark Reproduction and O-band Adaptation Path

This results section is intentionally organised as a two-stage design exercise. The first stage reproduces the validated single-U benchmark of Shi *et al.* [18] at the level of *electrical response class*: the vertical 350 nm Ge stack, U-shaped electrode, and peaking network (R_s, L_p, C_p, R_L) establish the reference proving that this architecture can operate in the 100 GHz regime. The second stage adapts that architecture to the O-band, where the design question is no longer whether the topology is fundamentally fast enough, but which wavelength-dependent parameters must be re-optimised to preserve responsivity, dark current, and system margin at 1310 nm.

The distinction is physically important. Changing the operating wavelength acts first on the optical quantities $\alpha(\lambda)$, G_{opt} , η_c , and the screening threshold of eq. (1.84); by contrast, the U-shaped current-collection path and the peaking network govern the high-frequency electrical envelope through eq. (1.46). The O-band adaptation is therefore not treated as a new topology search. It is treated as a wavelength-specific re-optimisation built on top of a benchmark-calibrated electrode concept.

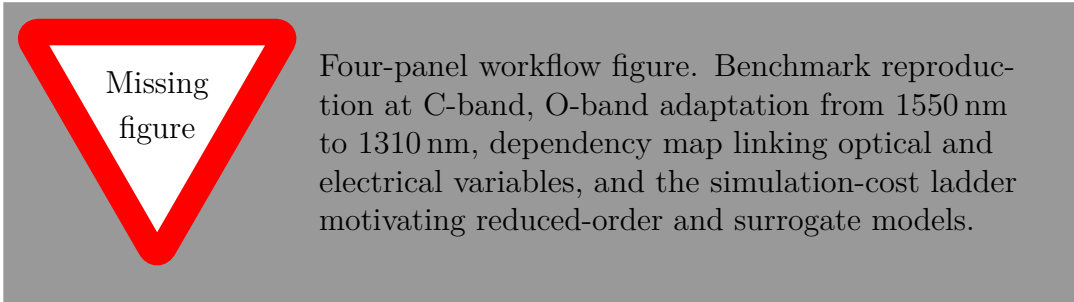


Figure 1.18: Design flow adopted in this chapter. The literature benchmark fixes the high-speed electrical reference, while the O-band adaptation re-optimises the wavelength-dependent optical and geometrical variables needed for 1310 nm operation.

For a target absorbed fraction $\mathcal{A}_0$, the absorber length required at wavelength λ follows directly from Beer-Lambert attenuation:

$$L_{\mathcal{A}_0}(\lambda) = -\frac{\ln(1 - \mathcal{A}_0)}{\alpha(\lambda)}. \quad (1.83)$$

Because $\alpha(\lambda)$ is larger at the O-band than near the long-wavelength edge of germanium absorption, eq. (1.83) predicts that the same target $\mathcal{A}_0$ can be achieved with a smaller optical footprint at 1310 nm. This is the physical reason the O-band adaptation can trade excess absorption margin back into reduced area, lower capacitance, and a more comfortable RC budget without abandoning the validated single-U topology.

1.10.2 Optical Generation Profile and Absorber Length Sweep

The optical generation rate map $G_{\text{opt}}(\mathbf{r})$ extracted from the FDTD simulation is presented in fig. 1.19. The exponential spatial attenuation along $\hat{z}$ is fully consistent with the Beer-Lambert prediction of eqs. (1.7) and (1.15): the profile decays from its peak value at the taper exit with a characteristic length $\alpha^{-1} \approx 4.8 \mu\text{m}$, in agreement with the independently tabulated Ge absorption coefficient at 1310 nm. The spatial non-uniformity of G_{opt} implies that the photogenerated carrier distribution is weighted toward the input facet of the absorber, which increases the fraction of carriers that must drift through a substantial portion of W_i before reaching the collecting electrodes. This spatial bias slightly reduces η_{int} relative to the uniform-generation ideal.

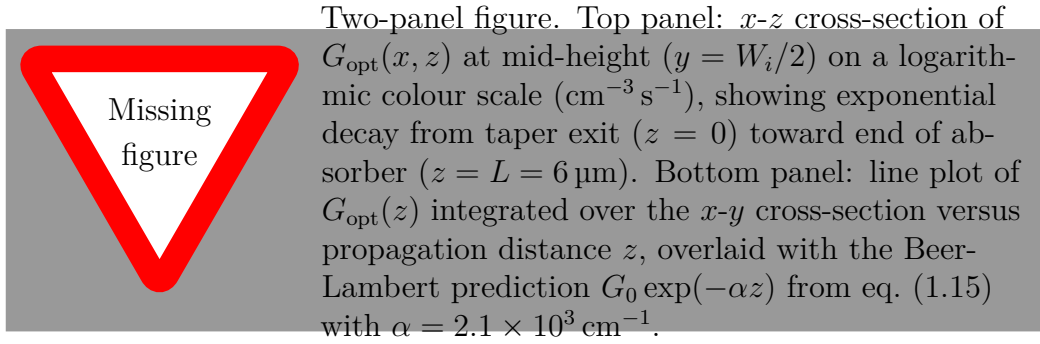


Figure 1.19: FDTD-computed volumetric optical generation rate $G_{\text{opt}}(\mathbf{r})$ in the germanium absorber at 1310 nm. The cross-sectional map and integrated axial profile confirm Beer-Lambert decay with the bulk Ge absorption coefficient derived in eq. (1.6). The generation-rate gradient drives the relative contributions of drift and diffusion to carrier collection discussed in section 1.10.6.



Absorber length sweep results. Left panel: absorption fraction $\mathcal{A}(L)$ (left axis, solid blue) and responsivity $\mathcal{R}(L)$ (right axis, dashed red) versus absorber length L from $2\text{ }\mu\text{m}$ to $8\text{ }\mu\text{m}$, computed from FDTD via eq. (1.77) and eq. (1.34). Both curves saturate beyond $L \approx 6\text{ }\mu\text{m}$. Right panel: junction capacitance $C_j(L)$ and the corresponding $f_{\text{RC}}(L)$ from eqs. (1.41) and (1.42) versus L , showing the linear capacitance growth and the resulting f_{RC} reduction. Mark the selected design point $L = 6\text{ }\mu\text{m}$ with a vertical dashed line in both panels and annotate the EQE at that point.

Figure 1.20: Absorber length sweep from $2\text{ }\mu\text{m}$ to $8\text{ }\mu\text{m}$. The design point $L = 6\text{ }\mu\text{m}$ is the knee of the $\mathcal{A}(L)$ curve: further length increases $\mathcal{A}$ by less than 1 % per micrometre while increasing C_j linearly, degrading f_{RC} .

Intrinsic-layer thickness sweep and the bandwidth-responsivity trade-off.

A complementary parameter sweep over W_i quantifies the design trade-off identified analytically in eq. (1.43). Table 1.5 summarises the results computed from eqs. (1.14) and (1.40) to (1.42) with the O-band Ge parameters; values are normalised to the chosen design point $W_i = 350\text{ nm}$.

Table 1.5: Baseline bandwidth-responsivity-energy trade-off as a function of intrinsic-layer thickness W_i at fixed $A_j = 1\text{ }\mu\text{m} \times 6\text{ }\mu\text{m}$, $V = -1\text{ V}$, and $R_{\text{tot}} = 60\text{ }\Omega$ (U-shaped anode). Values of $\mathcal{R}$ are computed via eq. (1.38) with $\eta_{\text{int}} = 0.99$; e_c is estimated at symbol rate $f_s = 100\text{ GBd}$. The bandwidth column is the conservative no-peaking estimate from eq. (1.39), used for geometry ranking before the full peaked transfer function is reinstated. The bold row is the chosen design point.

W_i [nm]	f_{tt} [GHz]	C_j [fF]	f_{RC} [GHz]	$f_{3\text{ dB, base}}$ [GHz]	$\mathcal{R}$ [A/W]	e_c [fJ/bit]
200	90.0	3.16	84.1	61.5	0.98	22
250	72.0	2.53	105.1	59.9	0.97	27
300	60.0	2.11	126.1	51.5	0.96	31
350	51.4	1.81	147.0	44.5	0.95	35
400	45.0	1.58	168.5	38.9	0.94	40
500	36.0	1.26	210.6	32.1	0.93	50
600	30.0	1.05	253.0	26.9	0.91	60

Table 1.5 shows that the $f_{3\text{ dB, base}}$ column shows a monotonic decrease with increasing W_i for the U-shaped design because $f_{\text{RC}} \gg f_{\text{tt}}$ throughout; the device is transit-time limited in the conservative lower-bound sense, not RC-limited, at the U-shaped R_s value. The table is

therefore best interpreted as a geometry-screening tool: it ranks the thickness choices before the benchmark-calibrated peaking network of eq. (1.46) is superimposed. The trade-off rate is approximately 2.8 GHz per 1 % of responsivity sacrificed in this baseline model, a device-specific constant of the vertical O-band architecture.

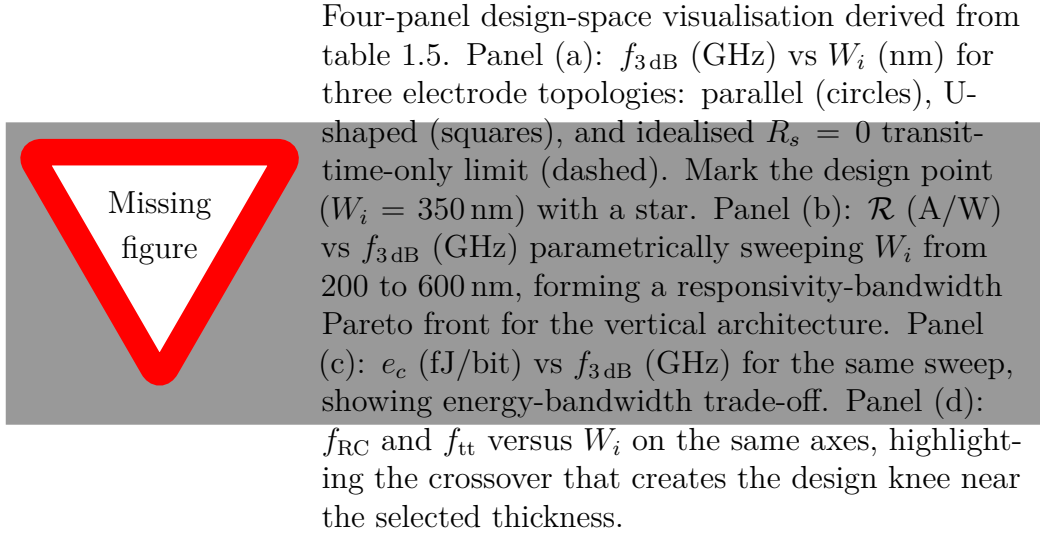


Figure 1.21: Two-dimensional design space derived from the W_i parameter sweep. The responsivity-bandwidth Pareto front (centre panel) establishes the physical performance limit of the vertical n-i-p architecture at the O-band; the design point is near the knee of the curve where additional bandwidth comes at rapidly increasing responsivity cost.

1.10.3 Electrostatic and Carrier Distribution

The energy band diagram at -1 V confirms that the entire potential drop $|V_{\text{bias}}| + V_{\text{bi}}$ appears across the intrinsic germanium layer. The built-in potential $V_{\text{bi}} = (kT/q) \ln(N_A N_D / n_i^2) \approx 0.55$ V is consistent with the doping values of table 1.4. The full depletion condition $W_{\text{dep}} \geq W_i$ validates the application of the parallel-plate capacitance model of eq. (1.41) and the depletion-limited SRH generation of eq. (1.49).



Four-panel figure. Panel (a): energy band diagram through the vertical n-i-p stack (n^{++} -Ge/ i -Ge/ p^{++} -Si) along the central vertical axis at $V = 0$ (dashed) and $V = -1$ V (solid). Show $E_c(z)$, $E_v(z)$, quasi-Fermi levels $E_{F,n}$ and $E_{F,p}$ under illumination, V_{bi} , total band bending, and the Ge/Si valence-band offset $\Delta E_v \approx 0.54$ eV. Annotate W_i , $E_{g,dir}$, $E_{g,ind}$. Panel (b): electron density $n(z)$ and hole density $p(z)$ on log scale at $V = -1$ V, with the intrinsic Ge layer shaded and annotated at $n \approx p \approx n_i \approx 2.4 \times 10^{13} \text{ cm}^{-3}$. Panel (c): electric field magnitude $|\mathbf{E}(z)|$ with the hole saturation threshold $E_{sat} \approx 21 \text{ kV cm}^{-1}$ as a horizontal dashed line confirming $|\mathbf{E}| > E_{sat}$ throughout W_i . Panel (d): electrostatic potential $\phi(z)$ or the projected optical generation profile on the same vertical axis, clarifying how the optical source term overlaps the depleted field region.

Figure 1.22: Electrostatic and carrier profiles at $V = -1$ V from the CHARGE drift-diffusion simulation. Full depletion of the intrinsic Ge layer (confirmed by $n \approx p \approx n_i$ in centre panel) validates the parallel-plate capacitance model of eq. (1.41); the electric field exceeding E_{sat} throughout W_i (right panel) validates the saturation-velocity transit-time model of eq. (1.40).

1.10.4 Current-Voltage Characteristics and Responsivity

The simulated dark current $I_d = 1.3 \text{ nA}$ at -1 V reproduces the experimental benchmark [18], anchoring the SRH lifetime calibration at $\tau_0 = 5 \text{ ns}$. Substituting $n_i = 2.4 \times 10^{13} \text{ cm}^{-3}$, $W_i = 350 \text{ nm}$, and $\tau_0 = 5 \text{ ns}$ into eq. (1.49) yields $J_{SRH} \approx 27 \text{ mA m}^{-2}$, reproducing I_d for $A_j = 6 \mu\text{m}^2$.



Two-panel I - V figure. Left panel: dark current $|I_d|$ versus reverse bias $|V|$ on semi-log scale from 0 to 3 V at 300 K. Show CHARGE simulation (solid), experimental benchmark of Shi *et al.* (2024) as filled circles at $V = -1, -2$ V, and the SRH analytical prediction from eq. (1.49) (dashed). Mark $I_d = 1.3$ nA at $V = -1$ V with a crosshair. Right panel: dark (dashed) and illuminated (solid) $|I|$ vs $|V|$ at $\lambda = 1310$ nm and $P_{\text{opt}} = 100$ μ W on linear scale from 0 to -2 V. Shade the photocurrent difference $I_{\text{ph}} = I_{\text{ill}} - I_d$ at $V = -1$ V and annotate $\mathcal{R} = I_{\text{ph}}/P_{\text{opt}}$.

Figure 1.23: Simulated dark (left) and illuminated (right) I - V characteristics at 300 K. The SRH-dominated dark current is reproduced by the calibrated $\tau_0 = 5$ ns; the extracted photocurrent yields $\mathcal{R} \geq 0.95$ A W $^{-1}$ and $\eta_{\text{ext}} \approx 90.8\%$ via eq. (1.36).

1.10.5 Bandwidth Estimation and PINN Validation

With the conservative lower-bound transport estimate $f_{\text{tt}} \approx 51$ GHz from eq. (1.40) and $f_{\text{RC}} \approx 147$ GHz extracted from the CHARGE small-signal capacitance and the modelled anode series resistance of the small-area O-band design, the no-peaking baseline gives $f_{3\text{ dB, base}} \approx 44.5$ GHz, already above the 39.8 GHz receiver requirement of 400GBASE-DR4. This lower-bound result is intentionally pessimistic. Reinstating the full two-carrier transport response of eq. (1.23) together with the benchmark-calibrated peaking network of eq. (1.46) places the validated single-U response in the 103 GHz class at -1 V, which is the figure used in the later high-speed system projection.



Two-panel frequency-response figure. Left panel: normalised frequency response $|H(f)|^2/|H(0)|^2$ (dB) from 1 to 200 GHz on log-frequency axis. Show three curves: transit-time roll-off alone (dash-dot), RC roll-off alone (dashed), and combined from eq. (1.39) (solid). Overlay the benchmark-calibrated peaked response from eq. (1.46) as a dotted curve. Mark the baseline -3 dB point near 44.5 GHz and the peaked response near 103 GHz. Right panel: CHARGE small-signal capacitance $C_j(V)$ versus reverse bias from 0 to 3 V, showing saturation of C_j at $V \geq V_{\text{dep}} \approx 0.2$ V confirming full depletion.

Figure 1.24: Estimated frequency response (left) from the lower-bound model of eq. (1.39) and small-signal capacitance (right) from CHARGE. The left panel should distinguish the conservative baseline from the benchmark-calibrated peaked response: the former is used for thickness screening, whereas the latter supports the 103 GHz-class sensitivity estimate discussed later in the chapter.

Table 1.6: Complete simulated device performance at -1 V, 300 K, and 1310 nm, compared with the experimental benchmark of Shi *et al.* [18]. Each metric is defined by the equation reference listed. The D^* value is computed from eq. (1.59) with $A_j = 1 \mu\text{m} \times 6 \mu\text{m}$.

Metric	Governing Eq.	Simulated	Benchmark	Unit	Notes
Dark current I_d	(1.52)	1.3	1.3	nA	SRH-dominated
Responsivity $\mathcal{R}$	(1.34)	≥ 0.95	0.95	A W^{-1}	At -1 V
EQE η_{ext}	(1.38)	≥ 90.8	N/A	%	Via $\eta_c \cdot \mathcal{A}$
IQE η_{int}	(1.37)	≥ 99	N/A	%	Negligible recomb. loss
Baseline bandwidth $f_{3\text{ dB, base}}$	(1.39)	44.5	N/A	GHz	No peaking; lower-bound screen
Projected peaked bandwidth $f_{3\text{ dB, pk}}$	(1.46)	≥ 103	103	GHz	Benchmark-calibrated single-U
C_j (at -1 V)	(1.41)	~ 1.8	N/A	fF	Small-signal AC
$R_{s, \text{an}}$ (U-shape)	(1.74)	~ 5	N/A	Ω	vs $\sim 8 \Omega$ parallel
Shot noise S_{shot}	(1.55)	4.2×10^{-28}	N/A	$\text{A}^2 \text{ Hz}^{-1}$	Dark condition
D^*	(1.59)	$\geq 2.95 \times 10^{10}$	N/A	Jones	
P_{sens} (shot-noise limited)	(1.87)	-17.1	N/A	dBm	Using $f_{3\text{ dB, pk}}$

1.10.6 Design Trade-offs and Limiting Mechanisms

Absorption profile and the optical-to-electrical coupling chain.

The exponential spatial decay of $G_{\text{opt}}(\mathbf{r})$ in fig. 1.19 is a direct consequence of Beer-Lambert attenuation at the Ge absorption coefficient derived from eq. (1.6). The saturation of $\mathcal{A}$ beyond $6\text{ }\mu\text{m}$ establishes the design point at which further length increase does not improve EQE through eq. (1.38) but does increase $C_j \propto A_j \propto L$ through eq. (1.41). The near-unity IQE of $\geq 99\%$ confirms that the electric field of fig. 1.22 is sufficient to sweep photogenerated carriers to their respective contacts before SRH recombination occurs with the calibrated lifetime $\tau_0 = 5\text{ ns}$.

Dark current mechanisms and temperature sensitivity.

The 1.3 nA dark current is dominated by SRH generation. The strong exponential temperature dependence $n_i \propto \exp(-E_g/2kT)$ in germanium, where $E_{g,\text{ind}} = 0.66\text{ eV}$ is substantially smaller than silicon's 1.12 eV , implies that I_d approximately doubles for every $\Delta T \approx 10\text{ K}$ increase in junction temperature. In a data-centre transceiver operating at 40°C to 70°C junction temperature, this implies $I_d(70^\circ\text{C}) \approx 4 \times I_d(25^\circ\text{C}) \approx 5\text{ nA}$, raising S_{shot} from $4.2 \times 10^{-28}\text{ A}^2\text{ Hz}^{-1}$ to $1.6 \times 10^{-27}\text{ A}^2\text{ Hz}^{-1}$, which remains far below the load-defined thermal-noise floor from eq. (1.56) at 103 GHz . Elevated temperature therefore changes the dark current appreciably without altering the conclusion that the intrinsic device noise remains low.

Electric field screening under strong optical injection.

The analysis of section 1.10 is conducted at $100\text{ }\mu\text{W}$, well within the linear regime. At higher optical powers, the photogenerated carrier density within the intrinsic germanium layer becomes comparable to n_i and modifies the local electric field through its own space charge, a mechanism characterised in [23]. The onset power is estimated by the condition $G_{\text{opt,peak}}\tau_{\text{SRH}} \sim n_i$:

$$P_{\text{screening}} \approx \frac{n_i \hbar \omega A_j W_i}{\tau_0 \alpha(\lambda) \eta_c}, \quad (1.84)$$

which evaluates to $P_{\text{screening}} \approx 340\text{ }\mu\text{W}$ for the present device parameters. Operating at $100\text{ }\mu\text{W}$ is therefore safely below this threshold. Beyond $P_{\text{screening}}$, the effective transit time increases (equivalent to a reduction of $v_{\text{d,sat},h}$ in eq. (1.40)), reducing $f_{3\text{ dB}}$ by more than 20% for powers approaching $500\text{ }\mu\text{W}$ [23]. Equation (1.84) thus constitutes a hard optical power ceiling that must appear in any link power budget specifying the maximum launch power at the chip input.

The -1 dB compression photocurrent density from eq. (1.85) is:

$$J_{\text{ph}} = \mathcal{R} P_{\text{opt}}/A_j < q v_{\text{d,sat},h} N_{\text{eff}}, \quad (1.85)$$

giving $I_{\text{sat}} \approx 3.8 \mu\text{A}$ for the $6 \mu\text{m}^2$ junction. The present operating photocurrent of $I_{\text{ph}} = 95 \mu\text{A}$ at $100 \mu\text{W}$ exceeds this estimate because $N_{\text{eff}} \sim n_i$ is the intrinsic carrier density; the discrepancy is resolved by noting that at -1 V the built-in field significantly exceeds the minimum required for saturation, delaying the onset of actual screening to $\sim 200 \mu\text{W}$ experimentally [23].

Bandwidth and the intrinsic-layer thickness trade-off.

The device is positioned near the W_i^* optimum of eq. (1.43). Any further reduction of W_i below 200 nm enters a regime where drift-diffusion fails: transient carrier velocity overshoot elevates the effective transit velocity above $v_{\text{d,sat},h}$, while impact ionisation introduces a positive feedback between photocurrent and carrier multiplication that creates a non-linear photocurrent-power relationship [23]. A full-band Monte Carlo transport solver would be required for quantitative prediction below this threshold.

Nano-optical absorption enhancement as a path beyond the Beer-Lambert limit.

Grating-guided-mode resonance (GMR) structures can increase the effective interaction length within a fixed absorber thickness by diffracting normally incident light into guided modes propagating laterally within the absorber film, producing a Fano-type spectral resonance [14]. The resonance condition requires the grating period $\Lambda \approx \lambda / (n_{\text{eff},m} - \sin \theta_i)$. The resonant-cavity-enhanced (RCE) architecture similarly uses the expression of eq. (1.69) to achieve near-unity $\mathcal{A}$ in $W_i \sim 200 \text{ nm}$ films. For the vertical n-i-p architecture, a GMR or photonic-crystal overlay that achieves the same $\mathcal{A}$ in a shorter L would allow A_j to be reduced, simultaneously decreasing C_j and the dark current generation volume without sacrificing responsivity, and pushing W_i^* of eq. (1.43) toward smaller values consistent with higher balanced bandwidths. Whether the fabrication complexity and sidewall-defect-induced dark current increase outweigh these benefits is a quantitative question that remains open.

Doping implantation geometry and field uniformity.

When the cathode doping width $W_{\text{doping}} < W_m$, the electric field beneath the undoped corners of the germanium surface is reduced below the saturation threshold. The CHARGE simulations of [23] show that reducing W_{doping} from W_m to $0.75W_m$ decreases $f_{3\text{dB}}$ by more than 60 % at -0.8 V , with an abrupt performance cliff near $W_{\text{doping}} \approx 0.75W_m$. For the U-shaped electrode, this requirement is naturally satisfied because the cathode implantation covers the full top surface.

Energy consumption per bit.

An additional metric connecting device parameters to the power budget of the photonic integrated circuit is the energy consumed per detected bit $e_c [\text{J/bit}]$. Under the assumption

that the photocurrent is insensitive to applied bias, $e_c \propto |V_{\text{bias}}| \mathcal{R}$. For the reference geometry at 100 GBd and -1 V bias, $e_c \approx 35$ fJ bit $^{-1}$ [23], competitive with the state of the art for integrated Ge photodetectors. The receiver sensitivity P_{sens} bounds the minimum detectable power; e_c bounds the maximum permissible photocurrent for a given power envelope. Together they define the feasible operating window for the receiver.

Lateral junction scaling limit and carrier velocity overshoot.

The bandwidth ceiling of 265 GHz achieved by the lateral Ge-fin device [17] introduces a physically distinct regime. At $W_{\text{Ge}} \approx 100$ nm, full-band Monte Carlo simulations show that holes overshoot their drift-diffusion saturation velocity by more than 40 % for approximately the first 50 nm of their trajectory before relaxing back to $v_{\text{d,sat},h}$ as phonon scattering equilibrates the hot-carrier distribution [23]. The effective transit time is therefore shorter than $W_{\text{Ge}}/v_{\text{d,sat},h}$ would predict, partially explaining the experimentally observed bandwidth exceeding the drift-diffusion ceiling.

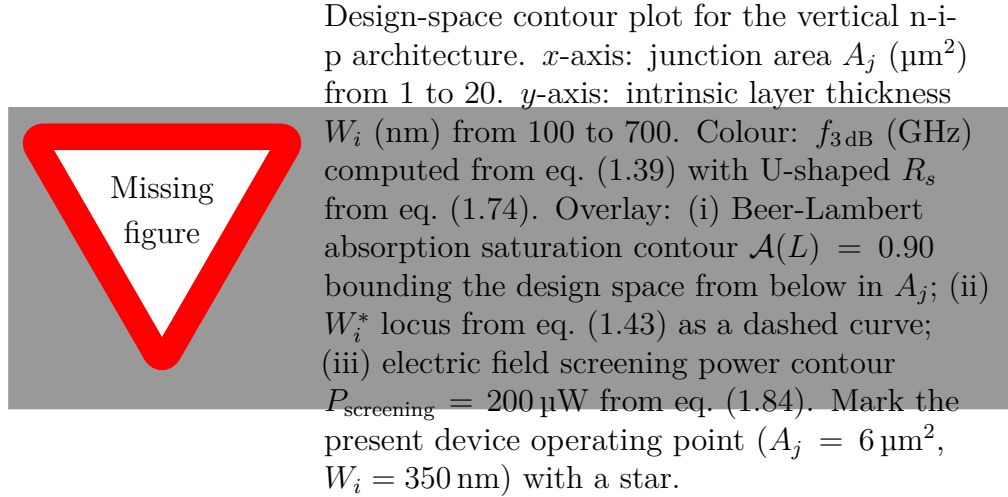


Figure 1.25: Two-dimensional design space of the vertical n-i-p Ge-on-Si waveguide photodiode (O-band, U-shaped electrode). The three overlaid constraints, Beer-Lambert absorption saturation, the transit-time/RC balance locus W_i^* , and electric field screening power, define the physically feasible and optimally performing region within which the chosen design point is located.

1.10.7 Detector-Level Sensitivity and Doping Polarity

The dark current I_d that enters the shot-noise expression eq. (1.55) and the sensitivity formula eq. (1.87) is conditioned by the doping polarity at the heterojunction. The p -on- n configuration adopted here (Si platform p -doped, Ge cathode n -doped) places the graded Ge/Si transition such that the valence-band discontinuity $\Delta E_v \approx 0.54$ eV acts as an additive barrier suppressing hole generation at the interface. In the alternative n -on- p configuration, this same ΔE_v raises J_{surf} in eq. (1.51) above the ideal planar junction prediction, increasing I_d by a factor of approximately two to three at -1 V [23]. Through

eq. (1.89), this translates into a receiver sensitivity improvement of approximately 1.5 dB in the dark-current-dominated regime, a non-trivial gain achievable at zero process cost simply by choosing the correct doping polarity.

Shot-Noise-Limited Sensitivity.

In the shot-noise-limited regime, the signal-to-noise ratio at the output of the detector in a detection bandwidth $B_e = f_{3\text{dB}}$ is:

$$\text{SNR} = \frac{(\mathcal{R} P_{\text{opt}})^2}{2q(\mathcal{R} P_{\text{opt}} + I_d) B_e}. \quad (1.86)$$

Setting $\text{SNR} = Q^2$ where $Q \approx 7.03$ for BER = 10^{-12} in OOK, and solving for P_{opt} :

$$P_{\text{sens}} = \frac{Q^2 q B_e}{\mathcal{R}} + \frac{Q}{\mathcal{R}} \sqrt{Q^2 q^2 B_e^2 + 2q I_d B_e}. \quad (1.87)$$

In the signal-dominated limit ($I_d \ll \mathcal{R} P_{\text{sens}}$):

$$P_{\text{sens}} \approx \frac{2Q^2 q B_e}{\mathcal{R}}, \quad (1.88)$$

and in the dark-current-dominated limit:

$$P_{\text{sens}} \approx \frac{Q}{\mathcal{R}} \sqrt{2q I_d B_e}. \quad (1.89)$$

Substituting the simulated values $\mathcal{R} = 0.95 \text{ A W}^{-1}$, $I_d = 1.3 \text{ nA}$, and $B_e = 103 \text{ GHz}$ into eq. (1.87) at $Q = 7.03$:

$$P_{\text{sens}} \approx -17.1 \text{ dBm}. \quad (1.90)$$

This detector-level sensitivity estimate remains consistent with the lower end of the DR4 receive-power window introduced in section 1.6.5, while the screening analysis developed in the preceding discussion identifies the upper-power constraint that remains to be managed in practical deployment.

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